

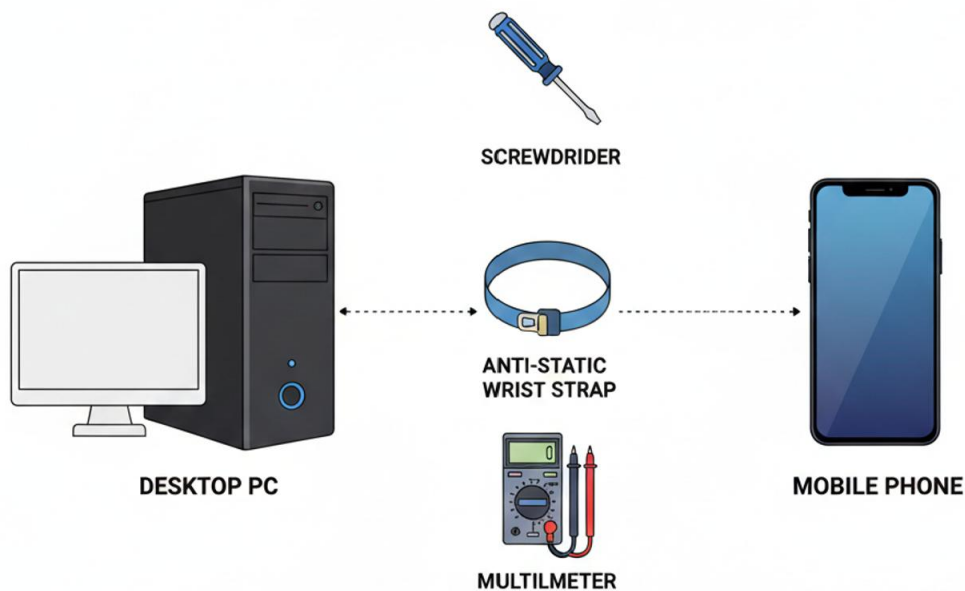
SS2 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF SS1 WORK

1.1 INTRODUCTION

Welcome to SS2. This term builds on the foundational knowledge from SS1, moving from basic component identification to advanced troubleshooting and mobile device repair. Before we proceed, we must ensure our understanding of SS1 concepts is solid.



1.2 RECALL OF KEY SS1 CONCEPTS

Last year, we focused on:

- **Safety Precautions:** Anti-static (ESD) rules, proper tool usage, and electrical safety.
- **Basic Tools:** Screwdriver sets (flat-head, Phillips, Torx), pliers, anti-static wrist straps, and multimeters.

- **Computer Components:** Identifying the main parts of a computer (CPU, Motherboard, RAM, Power Supply Unit (PSU), Hard Drive, Case).
- **System Assembly:** The process of putting these components together.
- **Introduction to Binary:** The concept of 0s and 1s as the language of computers.
- **Introduction to GSM:** What GSM (Global System for Mobile Communications) means and identifying basic phone parts (Screen, Battery, SIM slot, Charging port).

1.3 QUESTIONS AND ANSWERS (Q&A)

Q1: What is ESD, and why is it a major concern for a technician? A1: ESD stands for **Electrostatic Discharge**. It is the sudden flow of electricity between two electrically charged objects. The human body can build up thousands of volts of static electricity. While this is harmless to humans, this tiny "shock" is powerful enough to instantly destroy the delicate micro-circuitry on components like RAM, CPUs, and motherboards. This is why we use anti-static wrist straps and mats.

Q2: What is the primary function of the Power Supply Unit (PSU)? A2: The PSU has two main functions:

1. **Conversion:** It converts the high-voltage Alternating Current (AC) from the wall socket (e.g., 220V-240V in Nigeria) into the various low-voltage Direct Current (DC) voltages that computer components need (e.g., +12V, +5V, +3.3V).
2. **Distribution:** It distributes this DC power to all components (motherboard, hard drives, graphics card) via a series of dedicated cables and connectors.

Q3: What is the difference between RAM and a Hard Drive? A3:

- **RAM (Random Access Memory):** This is **volatile** (temporary) memory. It holds data for the applications that are *currently running*. When the computer is turned off, all data in the RAM is lost. It is extremely fast. Think of it as your *work desk*.
- **Hard Drive (HDD/SSD):** This is **non-volatile** (permanent) storage. It holds your operating system (Windows, macOS), all your files (photos, documents), and your

applications. When the computer is off, the data *stays* there. Think of it as your *filing cabinet*.

Q4: What is the "brain" of the computer, and where does it live? A4: The "brain" is the **CPU (Central Processing Unit)**. It performs all the calculations and executes instructions. It lives on the **Motherboard** in a special socket (e.g., LGA or AM4 socket) and is usually covered by a heat sink and fan for cooling.

Q5: What are the two main tools for diagnosing a GSM phone? A5:

1. **Digital Multimeter (DMM):** Used to check for voltage, continuity (broken lines), and resistance. It helps find faults on the circuit board.
2. **DC Power Supply:** Used to "force-boot" a phone without its battery. This helps determine if a fault is with the battery or the phone's power management circuit.

EVALUATION (WEEK 1)

1. What is the full meaning of ESD, and name two tools used to prevent it?
2. A client brings a PC that turns on, but the screen is blank and it makes a "beep...beep...beep" sound. What is this sound called, and what might it indicate? (Hint: SS1 work)
3. Differentiate between a Torx screwdriver and a Phillips screwdriver.
4. What is the difference between AC and DC power, and which one do computer components use?

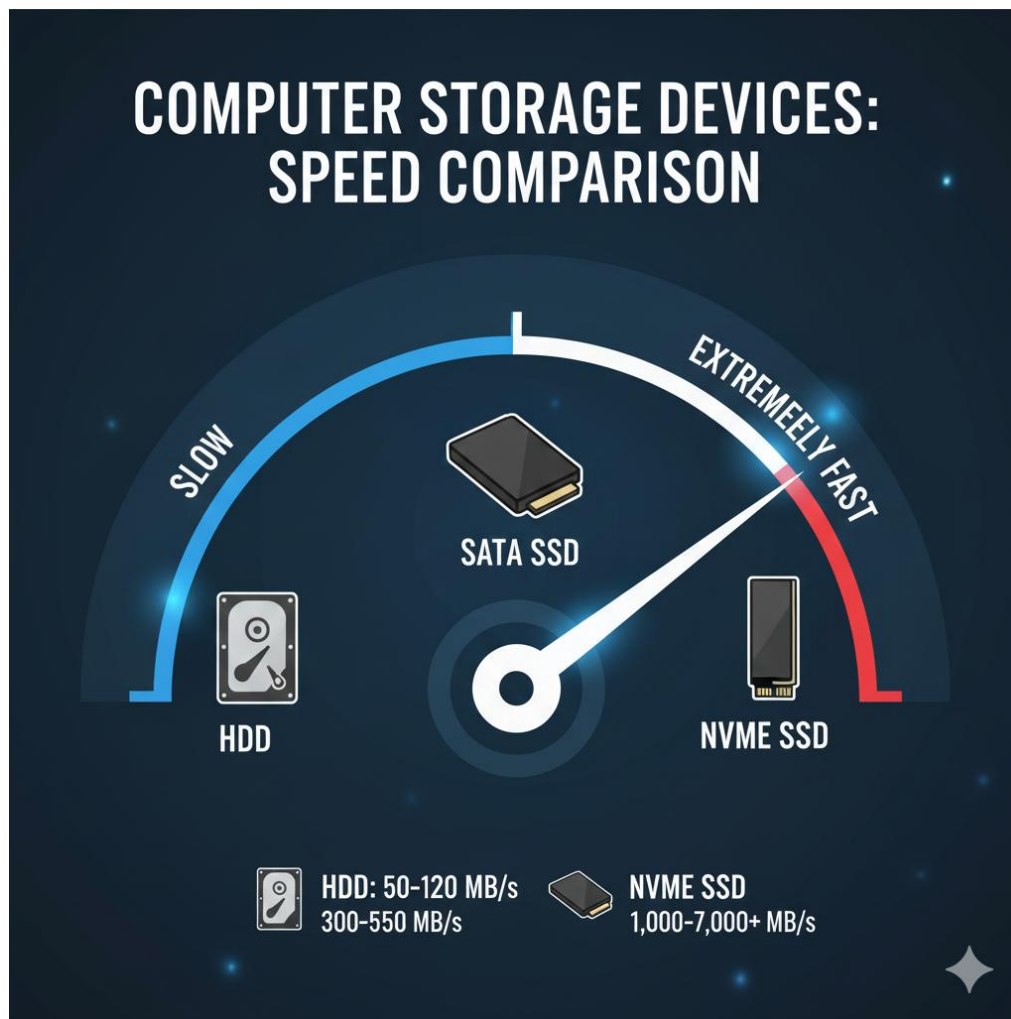
ASSIGNMENT (WEEK 1)

1. In a table format, list the six main components you would need to build a functional PC (excluding the case, keyboard, and mouse). For each component, write one sentence describing its function.

WEEK 2: COMPUTER STORAGE DEVICES

2.1 INTRODUCTION

Storage devices are a fundamental component of any computer system. They are non-volatile, meaning they retain data even when the power is turned off. We will explore the four main types of storage you will encounter.



2.2 HDD (HARD DISK DRIVE)

- **What it is:** The traditional, mechanical storage drive.

- **How it Works:** Data is stored magnetically on spinning, platter-like disks (similar to CD records). A mechanical "actuator arm" with a "read/write head" moves across these platters to find and change data.
- **Key Components:** Platter (the disk), Spindle (spins the platter), Actuator Arm, Read/Write Head.
- **Interfaces:**
 - **IDE (PATA):** Older, wide 40/80-pin ribbon cable. (Obsolete).
 - **SATA (Serial ATA):** Modern, small L-shaped connector for data and a separate one for power.
- **Pros:**
 - **High Capacity:** Available in very large sizes (e.g., 10TB, 20TB).
 - **Low Cost:** The cheapest storage per gigabyte (GB).
- **Cons:**
 - **Slow:** The mechanical moving parts create a bottleneck (slow boot times, slow file loading).
 - **Fragile:** Can be easily damaged by being dropped or shocked, leading to data loss.
 - **Noisy:** You can hear the platters spinning and the arm moving.
 - **Consumes more power** than an SSD.

2.3 SSD (SOLID STATE DRIVE)

- **What it is:** A modern, non-mechanical storage drive.
- **How it Works:** Uses **NAND flash memory chips** (similar to a flash drive or phone memory). It has no moving parts. Data is stored on and read from these chips by a "controller" chip.
- **Interfaces (Form Factors):**
 - **2.5-inch SATA:** Looks like a laptop HDD. Uses the same SATA connectors. Easiest upgrade for an old laptop or PC.
 - **M.2:** A small, "stick" like drive that plugs directly into a special slot on the motherboard. It's much faster.

- **NVMe (Non-Volatile Memory Express):** This is a *protocol* (language), not a connector. M.2 SSDs that use the NVMe protocol are the fastest, as they talk directly to the CPU via the PCIe bus.
- **Pros:**
 - **Extremely Fast:** 10x-50x faster than an HDD. (System boots in seconds, apps open instantly).
 - **Durable:** No moving parts, so it's shock-resistant and perfect for laptops.
 - **Silent:** Makes no noise.
 - **Low Power Consumption:** Improves laptop battery life.
- **Cons:**
 - **Higher Cost:** More expensive per gigabyte than an HDD.
 - **Limited Lifespan:** Flash memory can only be written to a certain number of times (though this is not a concern for most users).

2.4 OPTICAL DRIVES (CD/DVD/BLU-RAY)

- **What it is:** A drive that reads and writes data using a laser.
- **How it Works:** A laser reads "pits" (dents) and "lands" (flat areas) on the surface of a spinning disc (CD, DVD, or Blu-ray) as 0s and 1s.
- **Types:**
 - **CD-ROM/DVD-ROM:** Read-Only.
 - **CD-R/DVD-R:** Writable (can write *once*).
 - **CD-RW/DVD-RW:** Re-Writable (can write, erase, and write again).
 - **Blu-ray:** High-Definition (HD) format, holds much more data (25GB-50GB).
- **Pros:**
 - **Archival:** Good for long-term storage of data that won't be changed.
- **Cons:**
 - **Largely Obsolete:** Being replaced by flash drives and internet downloads.
 - **Very Slow:** Slower than all other storage types.
 - **Fragile:** Discs scratch easily.

2.5 FLASH DRIVES (USB DRIVES)

- **What it is:** A portable, small storage device using the same NAND flash memory as an SSD.
- **How it Works:** Connects to the computer via the **USB (Universal Serial Bus)** interface.
- **Pros:**
 - **Highly Portable:** Small, easy to carry.
 - **Universal:** Works on almost any device with a USB port.
 - **Durable:** No moving parts.
- **Cons:**
 - **Easy to Lose:** Their small size is also a disadvantage.
 - **Lower Lifespan/Speed:** Generally slower and less durable than a proper SSD.
 - **Security Risk:** A common way to spread computer viruses.

EVALUATION (WEEK 2)

1. What is the main physical difference between an HDD and an SSD?
2. A client wants to make their old, slow laptop "feel like new." What is the *single best* upgrade you would recommend, and why?
3. What do "SATA" and "NVMe" refer to?
4. Why are HDDs still popular, even though SSDs are much faster?
5. What does it mean when a drive is "non-volatile"?

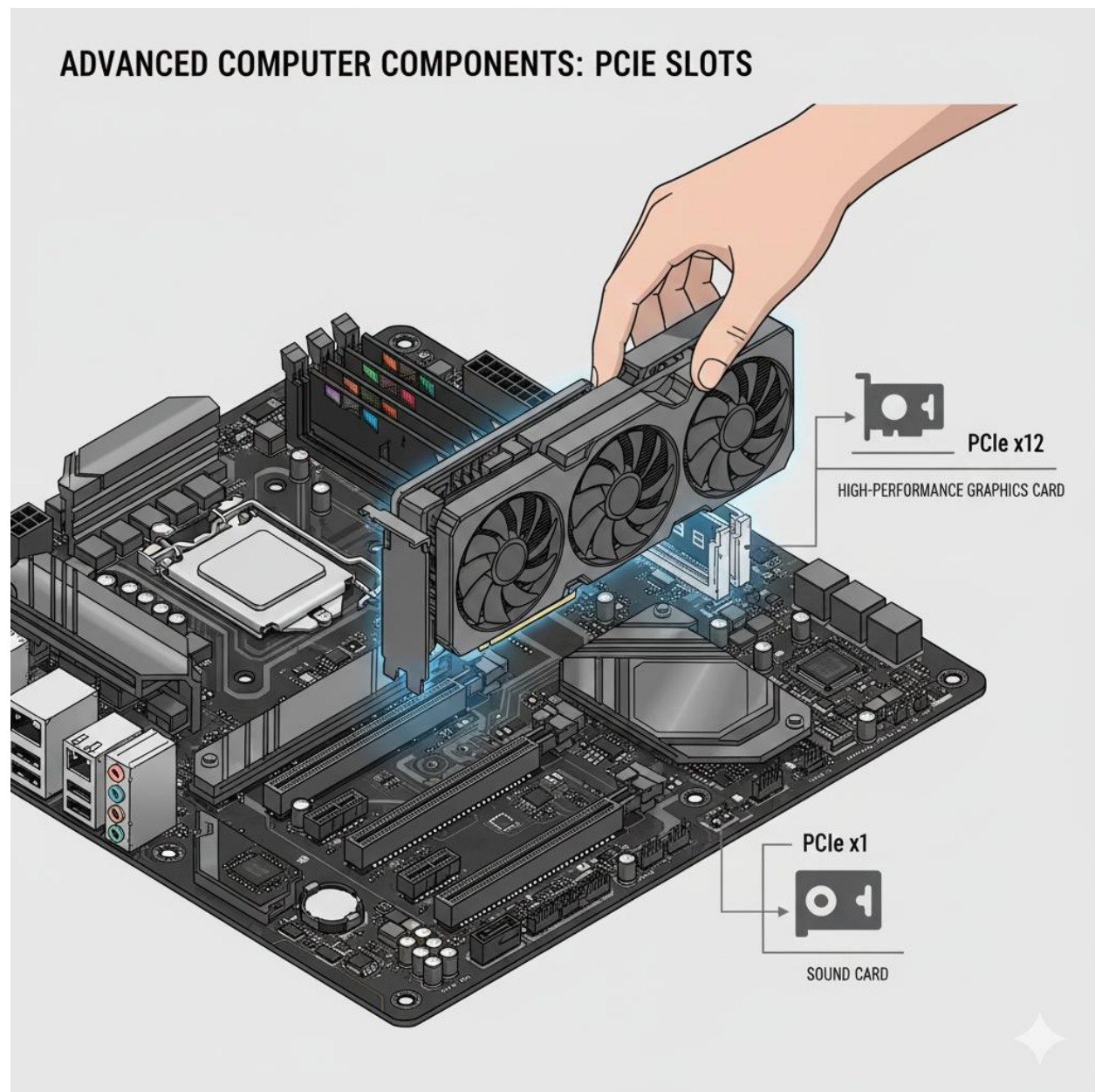
ASSIGNMENT (WEEK 2)

1. Go to a computer market or online store (e.g., Jumia/Konga).
2. Find the price for:
 - A 1TB HDD
 - A 1TB SATA SSD
 - A 1TB NVMe M.2 SSD
3. Create a table comparing their prices. Calculate the "cost per GB" for each (Price / 1000).
Write a one-paragraph conclusion on which one you would recommend for a new PC build and why.

WEEK 3: ADVANCED COMPUTER COMPONENTS

3.1 INTRODUCTION

While the CPU, RAM, and Motherboard are essential, "advanced" components provide specialized power for tasks like gaming, video editing, and professional audio work. These components connect to the motherboard via **expansion slots**.



3.2 EXPANSION SLOTS

- **What they are:** Sockets on the motherboard that allow you to "expand" the computer's capabilities by adding new "cards."
- **The Standard: PCIe (Peripheral Component Interconnect Express)**
 - This is the modern, high-speed "highway" that expansion cards use to talk to the CPU.
 - **Sizes:** You will see slots of different physical sizes:
 - **PCIe x1:** A tiny, short slot. Used for small cards like Wi-Fi adapters or simple sound cards.
 - **PCIe x4 / x8:** Medium-sized. Used for high-speed storage (NVMe adapter cards) or network cards.
 - **PCIe x16:** The longest, fastest slot. This is **exclusively** used for the Graphics Card (GPU).
- **Legacy Slots:**
 - **PCI:** An old, wide, white-coloured slot. Obsolete.

3.3 GRAPHICS CARD (GPU - GRAPHICS PROCESSING UNIT)

- **What it is:** A specialized computer-on-a-card with its own processor (the GPU) and its own RAM (VRAM). Its job is to process and render all the images, videos, and 3D scenes that you see on your monitor.
- **Types:**
 1. **Integrated Graphics (iGPU):**
 - The GPU is built directly into the CPU.
 - It has no VRAM; it "borrows" system RAM.
 - **Pros:** Cheap, low power.
 - **Cons:** Very weak. Only good for basic tasks (web browsing, office work).
 2. **Dedicated Graphics (dGPU):**
 - A separate, physical card that you buy and install into the PCIe x16 slot.
 - **Pros:** Extremely powerful. Essential for gaming, 3D modelling, video editing, and AI/machine learning.

- **Cons:** Expensive, consumes a lot of power (often needs its own power cables from the PSU), produces a lot of heat.
- **Key Components of a dGPU:**
 - **GPU Chip:** The actual "graphics processor" (e.g., NVIDIA GeForce, AMD Radeon).
 - **VRAM (Video RAM):** Its own ultra-fast RAM (e.g., 8GB, 12GB) for holding textures and images.
 - **Cooler:** A large heat sink and multiple fans to manage the extreme heat.
 - **Output Ports:** Where you plug in your monitor (HDMI, DisplayPort).

3.4 SOUND CARD

- **What it is:** A component that processes all audio input (microphone) and output (speakers, headphones).
- **Types:**
 1. **Onboard Sound:**
 - A small "codec" chip built into *every* modern motherboard.
 - The quality is "good enough" for 99% of users.
 2. **Dedicated Sound Card:**
 - A separate card installed in a PCIe x1 slot (or an external USB device).
 - **Why use one?**
 - **Audiophiles:** People who want the absolute highest, crystal-clear sound quality for listening to music.
 - **Music Producers:** Professionals who need to record high-quality audio from microphones and instruments with zero latency.
 - Provides better "shielding" from the electrical "noise" inside the PC case.

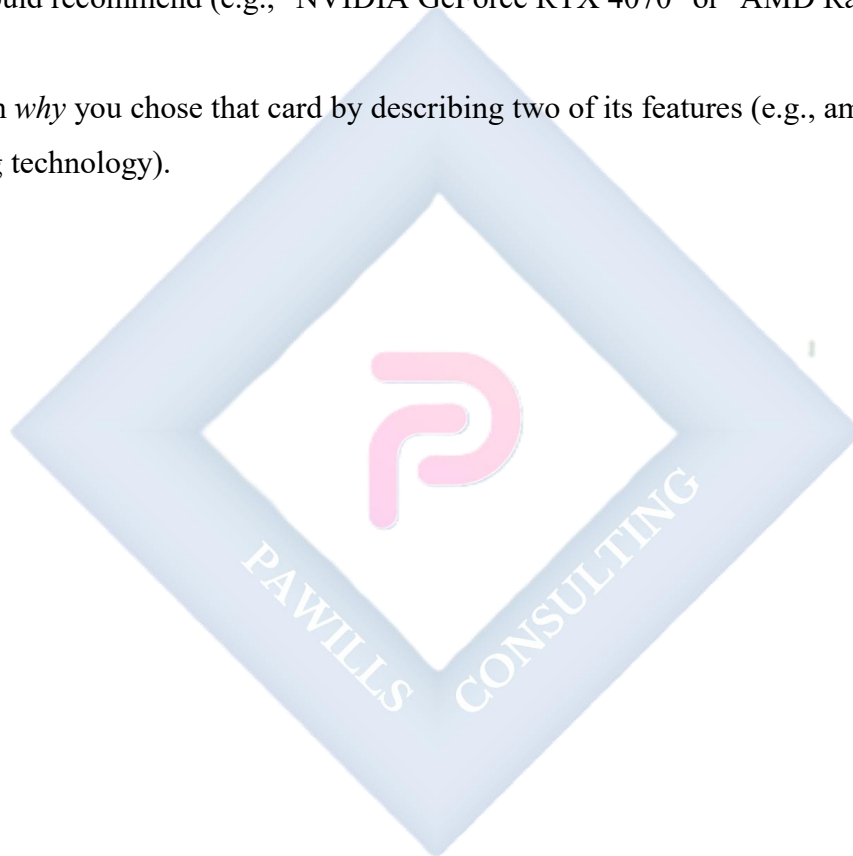
EVALUATION (WEEK 3)

1. What is the purpose of an expansion slot?
2. What is the name of the main, modern expansion slot, and which size is used for a graphics card?

3. Differentiate between "Integrated Graphics" and "Dedicated Graphics."
4. A client is a video editor who complains that "rendering videos takes forever." What is the first component you would suggest upgrading?
5. Why do most users *not* need a dedicated sound card? Who *does* need one?

ASSIGNMENT (WEEK 3)

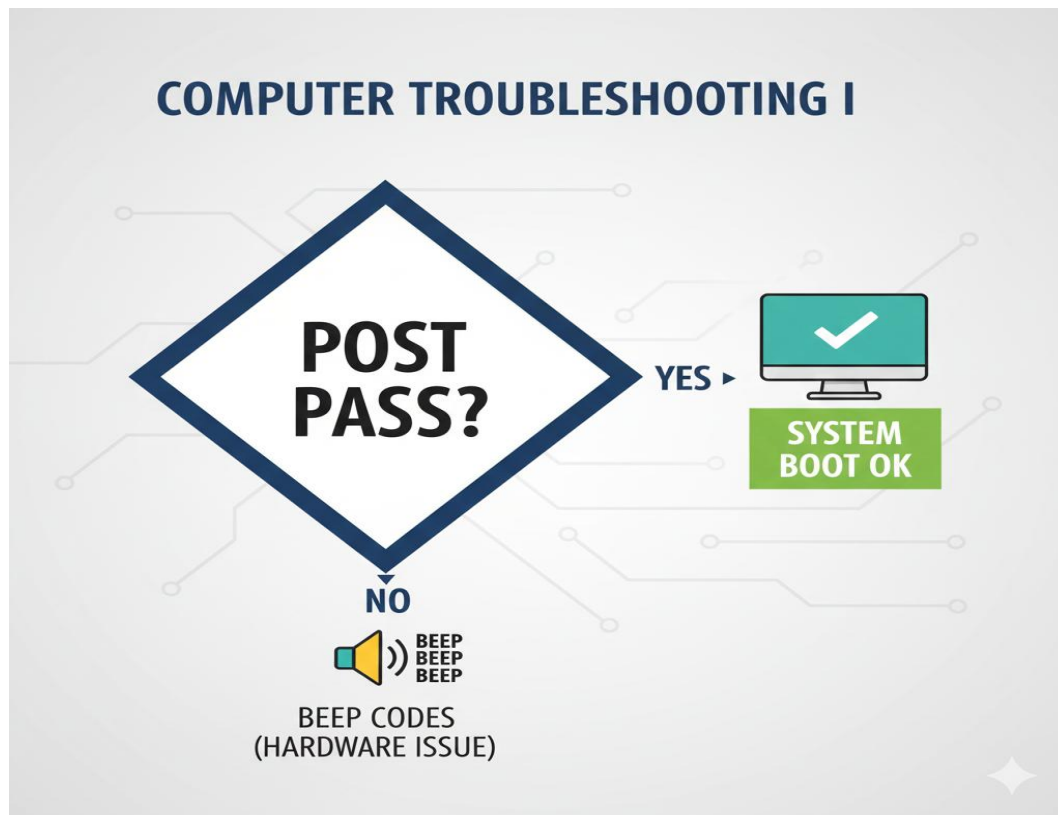
1. A client wants to build a high-end gaming PC. Research and list the *specific* components you would recommend (e.g., "NVIDIA GeForce RTX 4070" or "AMD Radeon RX 7800 XT").
2. Explain *why* you chose that card by describing two of its features (e.g., amount of VRAM, cooling technology).



WEEK 4: COMPUTER TROUBLESHOOTING I

4.1 INTRODUCTION

Troubleshooting is the logical, systematic process of finding and fixing a problem. It is the most important skill a technician can have. Randomly "trying things" is not troubleshooting.



4.2 THE 6-STEP DIAGNOSTIC PROCESS

Every technician should follow this process:

1. **Identify the Problem:** Question the user. What *exactly* is happening? What changed?
(e.g., "The computer won't turn on at all.")
2. **Establish a Theory (Hypothesize):** Based on the problem, what is the *most likely* cause?
(e.g., "No power at all. The most likely causes are a bad power cable, a bad wall socket, or a failed PSU.")

3. **Test the Theory:** Test your *easiest* and *most likely* theory first. (e.g., "Test a different power cable," "Test a different wall socket.")
4. **Implement a Solution:** Once the theory is confirmed (e.g., "The new power cable worked"), the problem is found.
5. **Verify Full Functionality:** The PC turns on. Now, does the OS load? Does the internet work? Ensure the fix didn't break something else.
6. **Document Findings:** Write down what the problem was and what the fix was. (This helps you and other technicians in the future).

4.3 POST (POWER-ON SELF-TEST)

- **What it is:** When you press the power button, the *first* thing the computer does is run a "roll call" of its essential hardware. This is the **POST**, controlled by the **BIOS/UEFI** (the firmware on the motherboard).
- **What it checks:**
 1. Is the CPU present and functional?
 2. Is RAM present and functional?
 3. Is the graphics card present?
 4. Is the keyboard present?
- **Pass:** If all these checks pass, you will hear **one single short beep**, and the computer will proceed to load the operating system.
- **Fail:** If any of these "essential" components fail the test, the computer will *stop* and *refuse to boot*. It will then report the error.

4.4 BEEP CODES

- **What they are:** If the POST fails and the video has not been initialized (so it can't show an error on-screen), the computer will use the small internal case speaker to report the error as a series of beeps.
- **This is a "secret code."** The meaning of the beeps depends on the **BIOS manufacturer** (e.g., AMI, Award, Phoenix, Dell, HP). You must listen to the pattern (e.g., long vs. short) and look up the code for that *specific* motherboard.

SAMPLE (AMI BIOS) BEEP CODES:

Beep Code	Meaning	Common Solution
1 Short Beep	DRAM Refresh Failure (Often a RAM or Motherboard issue)	Reseat the RAM. Test RAM one stick at a time.
2 Short Beeps	Parity Circuit Failure (RAM issue)	Reseat the RAM.
3 Short Beeps	Base 612K RAM Failure (RAM issue)	Reseat the RAM.
4 Short Beeps	System Timer Failure (Motherboard issue)	
5 Short Beeps	CPU Failure	Reseat the CPU. Check for bent pins.
1 Long, 3 Short	Video Card Failure	Reseat the graphics card. Check power cables.
1 Long, 8 Short	Display Test Failure	Check monitor cable. Reseat graphics card.
Continuous Beep	Power Supply Failure	Test or replace the PSU.

Key Takeaway: Multiple beeps (especially 1, 2, 3, or video-related) almost always mean **bad or incorrectly seated RAM or a bad graphics card**. The first step is *always* to **reseat** these components.

EVALUATION (WEEK 4)

1. A client says, "My computer is broken." What is the *first* step in the diagnostic process?
2. What is the POST, and when does it happen?
3. You power on a new PC build, and it gives "beep...beep...beep." What is the *first* component you should check?
4. Why is "Step 6: Document Findings" so important, even after the computer is fixed?
5. What is the difference between a POST failure and a boot failure?

ASSIGNMENT (WEEK 4)

1. Research the "beep codes" for a different BIOS type (e.g., "Award BIOS beep codes" or "Dell desktop beep codes").
2. Write down the meaning of:
 - 1 Long, 2 Short beeps
 - Continuous short beeps
 - 1 Long, 1 Short beep
3. (Make sure to state *which* BIOS type you researched).



WEEK 5: MIDTERM EXAMINATION

This week is for assessment of all topics covered in Weeks 1-4.



WEEK 6: COMPUTER TROUBLESHOOTING II

6.1 INTRODUCTION

In Week 4, we handled "No-POST" errors. This week, we handle what happens *after* the POST: the computer fails to load the Operating System (OS), or it crashes while running.



6.2 BOOT FAILURES

- **Symptom:** The POST completes (you hear one beep), but the OS (Windows) does not load.
- **Common Error Messages:**
 - "No bootable device found"
 - "BOOTMGR is missing" or "NTLDR is missing"
 - "Operating System not found"
 - A black screen with a blinking cursor (underscore).
- **Common Causes & Solutions:**
 1. **Wrong Boot Order in BIOS:**
 - **Cause:** The BIOS is set to boot from a USB drive or network, but the OS is on the HDD/SSD.
 - **Solution:** Enter the BIOS/UEFI setup (press F2, F10, or DEL on startup). Go to the "Boot" menu and ensure the "Windows Boot Manager" or the correct SSD/HDD is set as the #1 boot device.
 2. **Corrupt Boot Files:**
 - **Cause:** A bad Windows update, a virus, or an improper shutdown has corrupted the files needed to start Windows.
 - **Solution:** Boot from a Windows Installation USB drive. Select "Repair your computer" -> "Troubleshoot" -> "Startup Repair" or "Command Prompt" (to use commands like bootrec /fixmbr and bootrec /rebuildbcd).
 3. **Failed Storage Drive:**
 - **Cause:** The HDD or SSD that holds the OS has failed or its cable is loose.
 - **Solution:** Check if the drive is visible in the BIOS. If not, open the PC and check the SATA data and power cables. If it's still not seen, the drive is likely dead and must be replaced (and the OS reinstalled).

6.3 OVERHEATING

- **Symptom:** The PC works for a while (5 mins to an hour) and then *suddenly* shuts off, or it becomes extremely slow (this is "thermal throttling"). Fans are spinning very fast and loud.

- **Common Causes & Solutions:**

1. **Dust Buildup:**

- **Cause:** Dust has clogged the heat sinks (on the CPU and GPU) and case filters, acting like a blanket and trapping heat.
- **Solution:** Open the case. Use **compressed air** (or a data-vac) to blow the dust *out* of the heat sinks and fans.

2. **Failed/Failing Fans:**

- **Cause:** The CPU fan, GPU fan, or a case fan has died.
- **Solution:** Visually inspect that all fans are spinning. Replace any that are not.

3. **Dry Thermal Paste:**

- **Cause:** The **thermal paste** (a special substance that transfers heat from the CPU chip to the heat sink) has dried out after 3-5 years.
- **Solution:** A more advanced repair. You must remove the heat sink, clean *all* the old, dry paste off both the CPU and heat sink (using 99% isopropyl alcohol), and apply a new, tiny (rice-sized) dot of thermal paste before re-installing the heat sink. .

6.4 BLUE SCREEN OF DEATH (BSOD)

- **Symptom:** The computer suddenly stops, and the screen turns blue with an error message (in Windows).
- **What it is:** A "Stop Error." This is Windows's "emergency brake." It means a critical error has occurred, and the OS *must* shut down to prevent hardware damage.
- **What to do:**
 1. **Read the Stop Code:** Do not just restart. *Read the error* on the screen. It will say "Stop Code:" followed by a message (e.g., DRIVER_IRQL_NOT_LESS_OR_EQUAL).
 2. **Restart:** Restart the PC. If it happens again, you have a problem.
- **Common Causes & Solutions:**
 1. **Bad Drivers:**

- **Cause:** A newly installed driver (especially for a GPU, Wi-Fi, or printer) is conflicting with the OS.
- **Solution:** Boot into "Safe Mode." Uninstall the driver. Find an updated, stable driver from the manufacturer's website.

2. Faulty RAM:

- **Cause:** One of your RAM sticks is failing. This is a very common cause of random BSODs.
- **Solution:** Run "Windows Memory Diagnostic" (a built-in tool). To be certain, remove all RAM sticks. Test with *one stick* at a time. If the PC crashes with "Stick A" but not "Stick B," then Stick A is bad and must be replaced.

3. Corrupt System Files:

- **Cause:** A virus or bad update has damaged core Windows files.
- **Solution:** Open Command Prompt (as Admin) and run `sfc /scannow` (System File Checker). This will scan and repair corrupt files.

EVALUATION (WEEK 6)

1. What is the first thing you should check in the BIOS if you get a "No bootable device" error?
2. A client's PC shuts down *only* when they try to play a game. What is the most likely cause?
3. What is thermal paste, and what is its purpose?
4. What is a BSOD, and what is the most important piece of information on that screen?
5. What is the *easiest* way to test for a faulty RAM stick?

ASSIGNMENT (WEEK 6)

1. Research the "Stop Code" `IRQL_NOT_LESS_OR_EQUAL`.
2. Write a one-paragraph summary of what it means and what the three most common causes are.

WEEK 7: MIDTERM BREAK

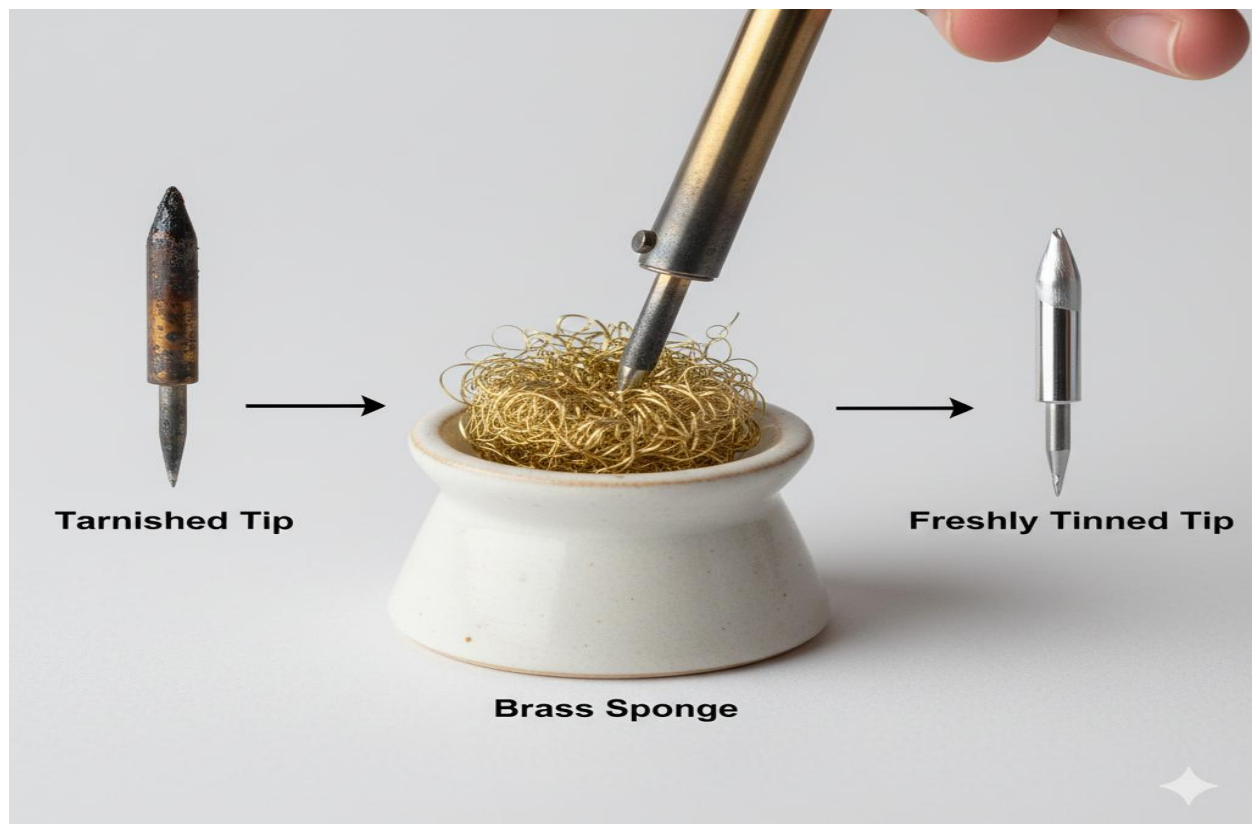
This is a break from instruction. Students are encouraged to practice troubleshooting on old/spare PCs if available.



WEEK 8: MAINTENANCE OF GSM TOOLS

8.1 INTRODUCTION

We now shift our focus to GSM (mobile phone) repair. Your tools are your "doctors," and they must be precise, clean, and well-maintained. Poorly maintained tools can damage a client's device.



8.2 SOLDERING IRON

- **Purpose:** To melt solder (a metal alloy) to join (or remove) components on a circuit board.
- **Maintenance:** The "tip" is the most important part.
 1. **Tinning the Tip:**
 - **What:** This is the process of coating the tip with a thin, fresh layer of solder.

- **Why:** It *prevents* oxidation (rusting) and allows heat to transfer efficiently to the joint.
- **How:** When the iron is hot, clean it, and immediately melt a small amount of fresh solder onto the tip. It should look shiny. Do this *before* you start and *before* you put it away.

2. Cleaning the Tip:

- **NEVER** use sandpaper or a file. This destroys the tip.
- **Brass Sponge:** The best method. Plunge the hot tip into the brass wool. It scrapes off oxidation without thermally shocking the tip.
- **Damp Sponge:** A wet sponge works, but it causes "thermal shock" (rapid cooling/heating) which can shorten the tip's life.

3. Temperature: Do not use a temperature that is hotter than you need. Excessive heat will cause the tip to oxidize and fail quickly.

• Safety:

- **Always** use a proper stand. Never lay a hot iron on the bench.
- Work in a ventilated area. Solder "flux" (the chemical core) produces fumes that are bad to inhale.
- Turn it off when you are not using it.

8.3 HOT AIR REWORK STATION (HOT AIR GUN)

- **Purpose:** Used to remove and replace **SMD (Surface Mount Devices)** and **ICs (Integrated Circuits)** (the black chips) by heating the *entire* area.
- **Maintenance:**
 1. **Nozzle Care:** Use the correct nozzle for the job. Keep them clean.
 2. **Handle with Care:** The heating element inside is fragile. Do not drop the wand.
 3. **Cool-Down Cycle:** This is critical. *Never* just unplug a hot air station. When you turn it "off," the *fan* will continue to run (even as the heat turns off) to cool the heating element. You must wait for this cycle to finish before unplugging it, or you will destroy the element.
- **Safety:**

- **Extremely Hot:** This tool outputs air at 300°C - 450°C. It can *instantly* melt plastic connectors and burn the PCB (turn it brown).
- **Shielding:** Use Kapton tape (heat-resistant yellow tape) or metal shields to protect nearby plastic components.
- **Airflow:** Use *just enough* airflow. Too much can blow tiny resistors and capacitors right off the board.

8.4 DIGITAL MULTIMETER (DMM)

- **Purpose:** Your main diagnostic tool. It measures Voltage (V), Resistance (Ω), and Continuity (beep test).
- **Maintenance:**
 1. **Batteries:** If you see a "low battery" icon, replace it immediately. A low battery will give you *inaccurate* readings.
 2. **Fuses:** The DMM has an internal fuse to protect it. If you (for example) try to measure Amps incorrectly, you will blow the fuse. The DMM will *appear* to work, but some functions won't. Always have spare fuses.
 3. **Probes (Leads):** These are the red and black wires. Check them for breaks or cracks. A damaged probe will give false readings.
- **Safety:**
 - **Turn it OFF:** When you are done, turn the dial to "OFF" to save the battery.
 - **Know Your Function:** *Always* set the dial to the correct function (Volts, Ohms, etc.) *before* you touch the probes to the circuit.
 - **NEVER** test for Resistance (Ω) or Continuity on a **LIVE (powered-on)** circuit. This will give wrong readings and can damage the meter.
 - **NEVER** touch the metal tips of the probes together when testing Voltage.

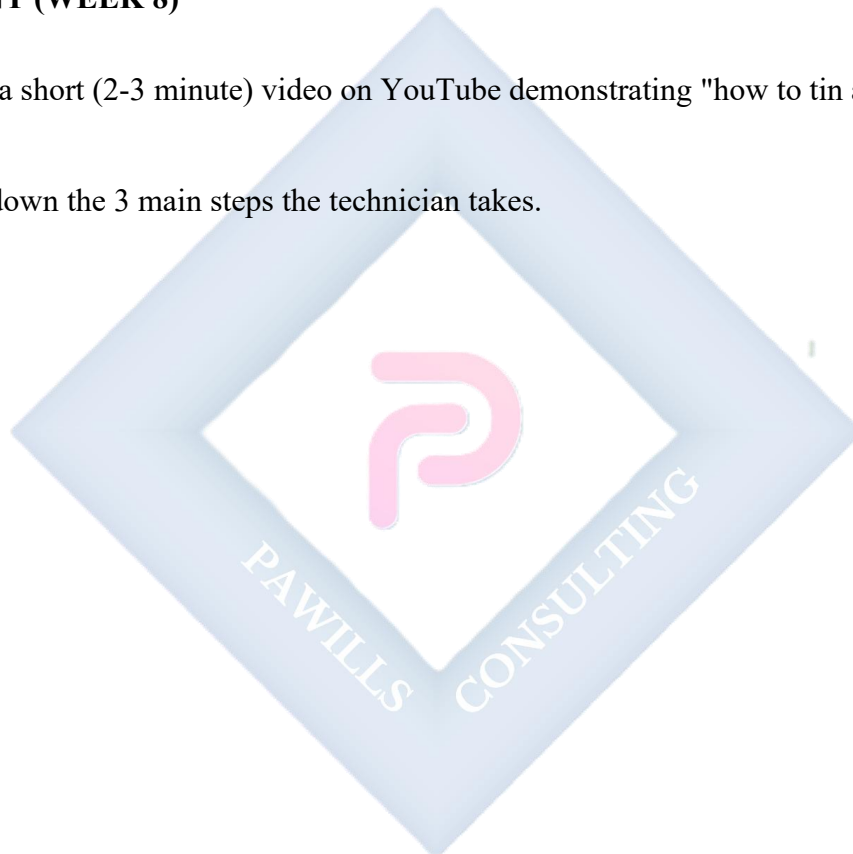
EVALUATION (WEEK 8)

1. What is "tinning" a soldering iron tip, and why is it important?
2. What is the *best* way to clean a hot soldering iron tip, and what method is "okay" but causes thermal shock?

3. What is the *most important* step to take after you finish using a hot air rework station (before unplugging it)?
4. A technician's multimeter is giving readings that are "all over the place." What is the most likely, simple cause?
5. What is Kapton tape used for in GSM repair?

ASSIGNMENT (WEEK 8)

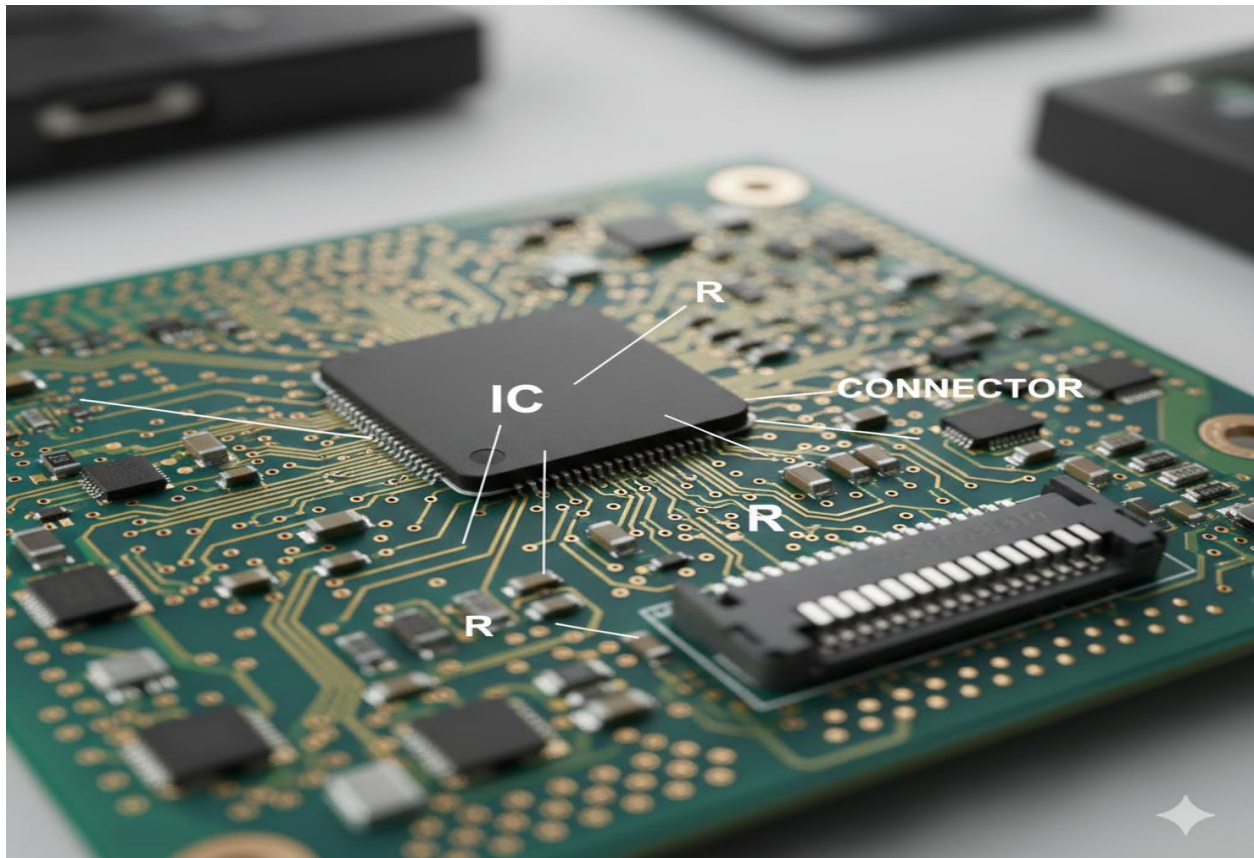
1. Watch a short (2-3 minute) video on YouTube demonstrating "how to tin a soldering iron."
2. Write down the 3 main steps the technician takes.



WEEK 9: GSM CIRCUIT BOARDS

9.1 INTRODUCTION

A GSM phone's circuit board (called a **PCB - Printed Circuit Board**) is a miniature city. All the components are "surface-mounted," meaning they are soldered to pads on the *surface* of the board, not put through holes. These are called **SMD (Surface Mount Devices)**.



9.2 IDENTIFICATION OF KEY COMPONENTS (VISUAL)

- **1. ICs (Integrated Circuits) / "Chips":**
 - **What they are:** The "brains" and "organs" of the phone. These are the large (and small) black, rectangular chips.
 - **How to identify:** Usually the largest components, covered in black epoxy, often with a metal shield over them.
 - **Examples:**

- **CPU/SoC (System on a Chip):** The main processor. The biggest chip, often under the memory.
- **PMIC (Power Management IC):** The "PSU" of the phone. Manages the battery, charging, and distributes power to all other chips. A *very* common point of failure.
- **Flash Memory (Storage):** The phone's "hard drive" (e.g., 64GB, 128GB).
- **RF (Radio) IC:** Manages the network/SIM signal.
- **Audio IC:** Manages the speaker and microphone.
- **2. Resistors (R):**
 - **What they do:** Resist the flow of electricity.
 - **How to identify:** Tiny, *black* rectangles with metal ends. They are the most common component on any board.
 - **On a schematic:** Labelled "R" (e.g., R2001).
- **3. Capacitors (C):**
 - **What they do:** Store and release energy. They act as filters, "cleaning" the power.
 - **How to identify:** Tiny, *light-brown, beige, or grey* rectangles with metal ends.
 - **Note:** These are *very* easily "shorted to ground" and are a common cause of faults.
 - **On a schematic:** Labelled "C" (e.g., C1004).
- **4. Inductors (Coils) (L):**
 - **What they do:** Store energy in a magnetic field. Also used for filtering power.
 - **How to identify:** Often look like larger, grey/black blocks, or small wire-wound "coils."
 - **On a schematic:** Labelled "L" (e.g., L3002).
- **5. Diodes (D) / LEDs:**
 - **What they do:** Allow electricity to flow in *one direction* only.
 - **How to identify:** Usually black, with a small line or symbol indicating the direction. (An LED - Light Emitting Diode - is a diode that lights up, like your camera flash).
 - **On a schematic:** Labelled "D" (e.g., D001).
- **6. Connectors:**
 - **What they do:** Connect other parts to the main board.

- **How to identify:** Obvious plastic or metal sockets.
- **Examples:** Battery connector, LCD/Screen connector, Charging port connector.

EVALUATION (WEEK 9)

1. What is the difference between an SMD component and a "through-hole" component?
2. What is the PMIC, and why is it so important?
3. I see a tiny component on a board. How can I visually guess if it is a resistor or a capacitor?
4. What is the main function of an Inductor (Coil)?
5. What do "R," "C," and "L" stand for on a circuit diagram?

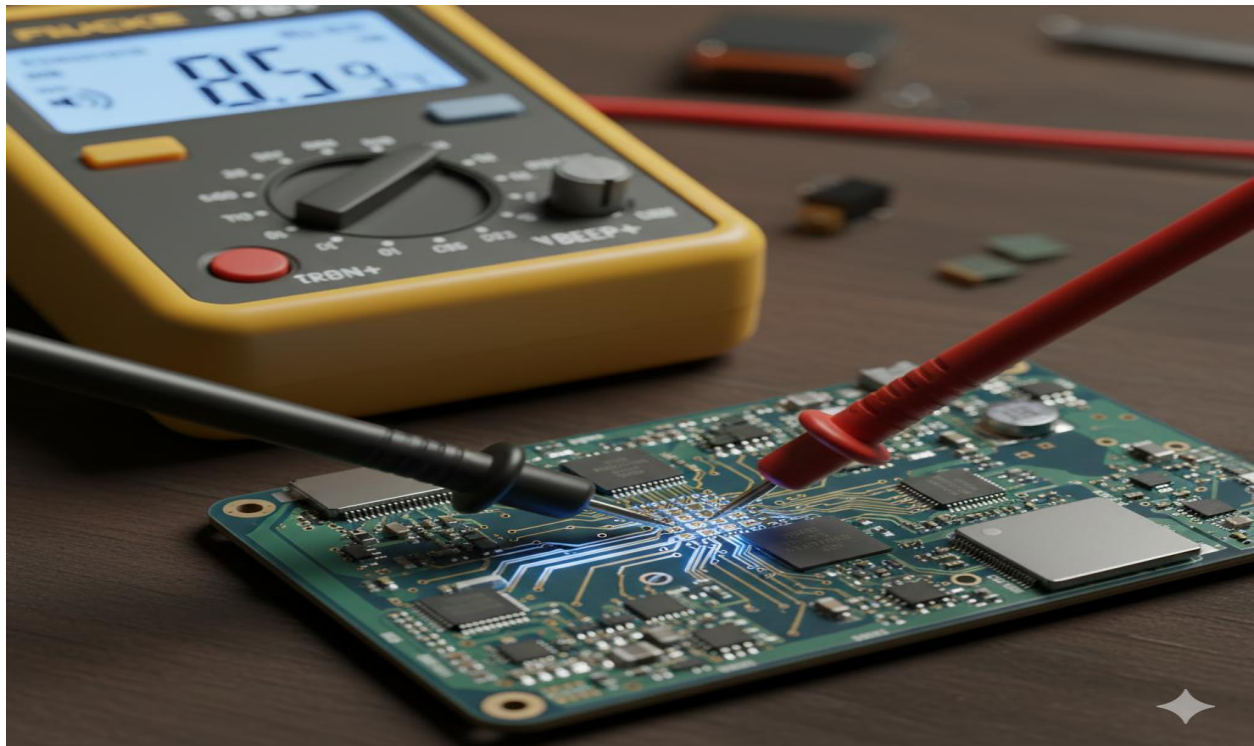
ASSIGNMENT (WEEK 9)

1. Find an old, dead piece of electronics (an old remote, a dead phone, a calculator).
2. Open it up and look at the PCB.
3. Try to identify and point out (or sketch) the following:
 - An IC
 - A Resistor
 - A Capacitor
 - A Connector

WEEK 10: GSM CIRCUIT TRACING

10.1 INTRODUCTION

"Circuit Tracing" is the practical skill of using your multimeter to "follow" the electrical paths (traces) on a PCB to find a fault. The two most common faults are **Short Circuits** and **Open Circuits**.



10.2 USING THE MULTIMETER (DMM) FOR TRACING

You will use three main functions:

- **1. Continuity Mode (Beep Test):**
 - **What it does:** The DMM "beeps" if the two probes are connected with very low resistance.
 - **How to use:** Set the dial to the "Continuity" symbol (looks like a sound wave).
 - **This is your #1 test.**
 - **Purpose:** To check if a trace is "open" (broken) or "shorted."

- **2. Voltage Mode (DC Volts):**
 - **What it does:** Measures the presence and level of (DC) voltage.
 - **How to use:** Set the dial to "V" with a straight line (DCV). **This test MUST be done with the phone powered ON (or plugged into the DC supply).**
 - **Purpose:** To check if power is reaching a component. (e.g., "Is 1.8V reaching the camera connector?").
- **3. Resistance Mode (Ohms Ω):**
 - **What it does:** Measures the amount of resistance.
 - **How to use:** Set the dial to " Ω ". **This test MUST be done with the phone powered OFF.**
 - **Purpose:** To check if a component (like a resistor) has the correct value.

10.3 FINDING FAULTS

A. OPEN CIRCUIT (THE "OPEN")

- **The Fault:** A trace (the line) on the PCB is broken. A component is "not connected."
- **Symptom:** A function just "doesn't work" (e.g., the earpiece speaker is silent).
- **How to find:**
 1. Use **Continuity Mode** (Beep Test).
 2. Place one probe on the connector pin for the speaker.
 3. Place the other probe on the component that the trace *should* go to (you need a schematic/diagram for this).
 4. **Result: NO BEEP.** This means the line is broken.
 - **Solution:** You must run a "jumper wire" — a tiny, insulated wire that you solder from the first point to the second, "jumping" over the break.

B. SHORT CIRCUIT (THE "SHORT")

- **The Fault:** A component or trace that should be "hot" (have voltage) is touching "Ground" (the negative, or the metal frame). This is the *most common* fault in dead phones.

- **Symptom:** The phone is "dead," won't power on, or gets extremely hot. The battery drains in minutes.
- **How to find (The "Beep" Test):**
 1. Use **Continuity Mode**.
 2. "Ground" is the entire metal shielding, the SIM tray, or the negative terminal of the battery.
 3. Place your **Black probe** on "Ground."
 4. Place your **Red probe** and "poke" around the components (e.g., on both sides of the capacitors).
 5. **Result:** You are looking for a component that **beeps on BOTH sides**. (A capacitor should *never* beep on both sides). If it does, that capacitor (or the line it's on) is **shorted to ground**.
 - **Solution:** That component (or another one on the same line) is faulty and must be removed using the hot air gun.

EVALUATION (WEEK 10)

1. What are the two most common faults in GSM repair?
2. What is the *first* multimeter mode you should use to check a dead phone?
3. When testing for a "Short," what does it mean if a capacitor "beeps on both sides"?
4. When do you use "Voltage Mode," and what is the main safety rule?
5. What is the solution for an "Open Circuit"?

ASSIGNMENT (WEEK 10)

1. Research "how to find a short circuit on a phone board."
2. Besides using a multimeter, what is another common method technicians use to *visually* find the exact shorted component? (Hint: It involves "freeze spray" or "rosin"). Write a short paragraph describing this method.

WEEK 11: REVISION

This week is for a comprehensive review of all topics covered in the First Term, from PC components to GSM troubleshooting.

- **PC Hardware (Weeks 2-3):**

- Storage: HDD (mechanical, slow), SSD (flash, fast), M.2/NVMe (fastest).
- Expansion: PCIe slots (x16, x1), GPU (integrated vs. dedicated), Sound Cards.

- **PC Troubleshooting (Weeks 4, 6):**

- POST & Beep Codes: The "roll call" of hardware. Beeps often mean RAM or GPU.
- 6-Step Process: The logical method.
- Boot Failures: "No boot device," "BOOTMGR missing." Check BIOS, cables, or repair OS.
- Overheating: Caused by dust, bad fans, dry thermal paste.
- BSOD: Stop code, bad drivers, bad RAM.

- **GSM Repair (Weeks 8-10):**

- Tools: "Tinning" the soldering iron, "cool-down" of the hot air gun, battery/fuse care for the DMM.
- PCB Components: ICs (PMIC), Resistors (black), Capacitors (brown/beige), Coils (L).
- Tracing: Continuity "beep test" is #1. Open (no beep) vs. Short (beep to ground).

SS2 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

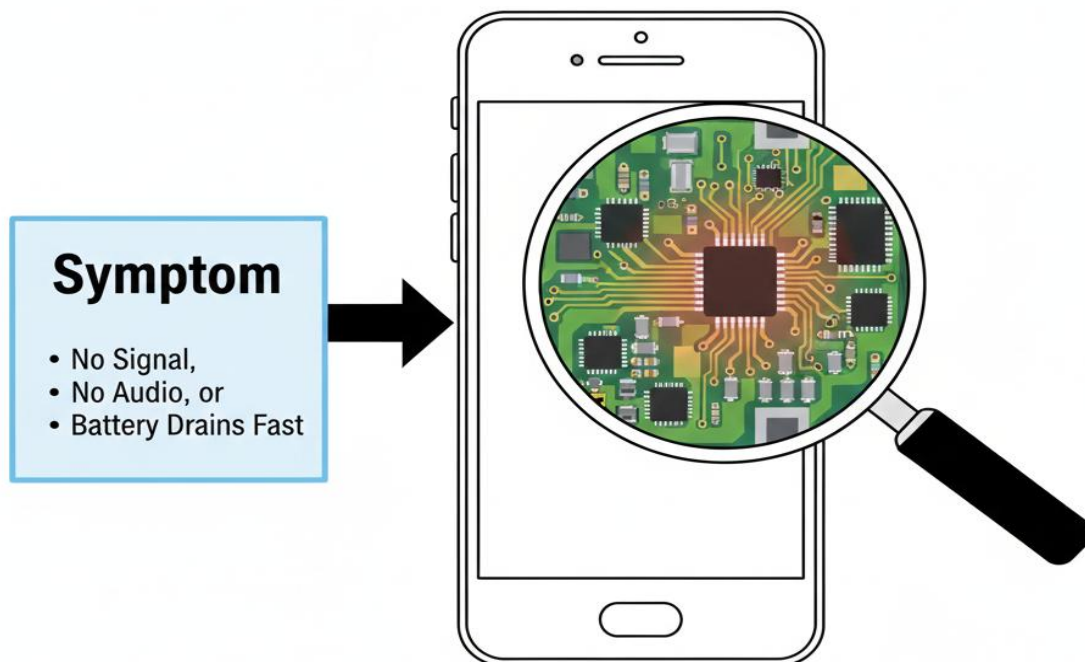
SECOND TERM

WEEK 1: REVIEW OF GSM HARDWARE

1.1 INTRODUCTION

Welcome to the Second Term. Having mastered advanced PC hardware, we now dive deep into the world of GSM (Global System for Mobile Communications) repair. This is a high-demand, high-skill field.

REVIEW OF GSM HARDWARE



1.2 RECAP OF SS1 GSM FAULTS (SYMPTOM-LEVEL)

In SS1, we identified basic, "user-level" faults. These are the *symptoms* that the client reports.

- **Physical Damage:** Cracked screen, broken back glass, damaged charging port.
- **Power Issues:** "Phone is dead," "Battery drains too fast," "Phone won't charge."
- **Audio Issues:** "I can't hear them," "They can't hear me," "Loudspeaker not working."
- **Software Issues:** "Phone is stuck on logo (boot loop)," "Phone is slow," "I forgot my password."
- **Connectivity Issues:** "No network," "Wi-Fi won't turn on."

1.3 THE SS2 GOAL: ROOT CAUSE ANALYSIS

This term, we move beyond the *symptom* to find the *root cause* at the **circuit board level**.

- **Example 1:**
 - **Symptom:** "Phone won't charge."
 - **SS1 Diagnosis:** The charging port is dirty or broken. The battery is bad.
 - **SS2 Diagnosis:** The charging port is fine, but the **Charging IC** on the motherboard has failed. Or, a capacitor on the **VBUS** line is shorted to ground.
- **Example 2:**
 - **Symptom:** "No Network."
 - **SS1 Diagnosis:** The SIM card is bad. The phone is in airplane mode.
 - **SS2 Diagnosis:** The **antenna coaxial cable** was damaged during a screen replacement, or the **RF (Radio Frequency) IC** has a cold solder joint.
- **Example 3:**
 - **Symptom:** "Phone is dead."
 - **SS1 Diagnosis:** The battery is dead.
 - **SS2 Diagnosis:** The **PMIC (Power Management IC)** is shorted, and the phone is drawing 1A on the DC power supply, but not booting.

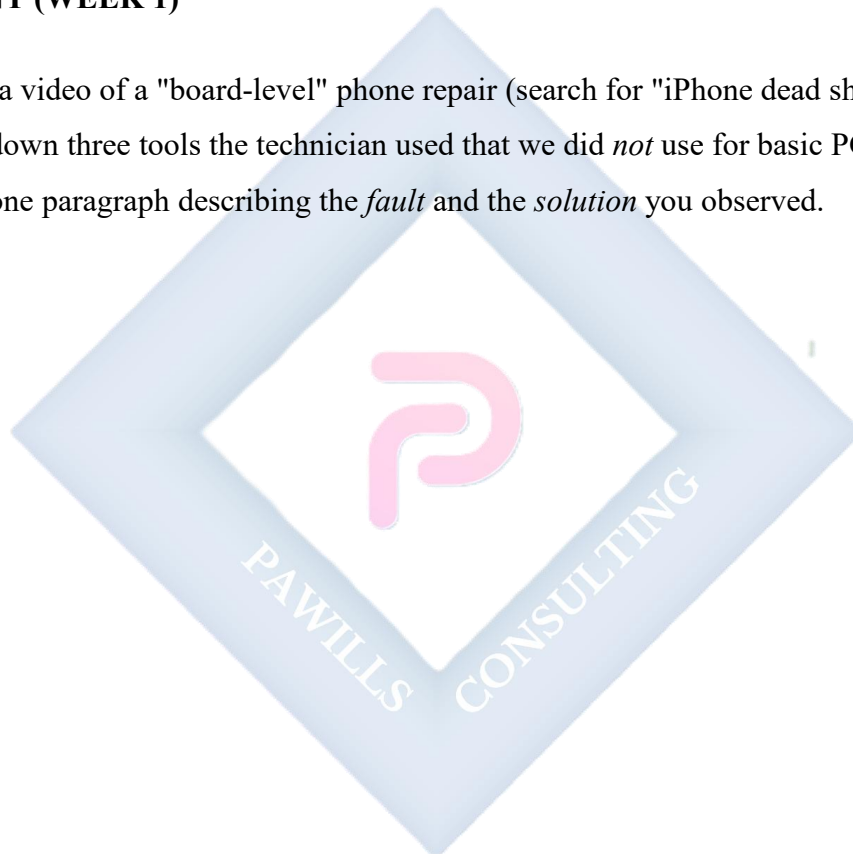
This term, we learn to use advanced tools (multimeter, hot air, DC supply, software boxes) to diagnose and fix these board-level faults.

EVALUATION (WEEK 1)

1. A client brings a phone with a "No Network" error. List one *SSI-level* check and one *SS2-level* (board) check you would perform.
2. What is the difference between a "symptom" and a "root cause" in GSM repair?
3. A client's phone is "stuck on the logo." Is this more likely a hardware or software fault?
4. What does PMIC stand for, and what is its general function?

ASSIGNMENT (WEEK 1)

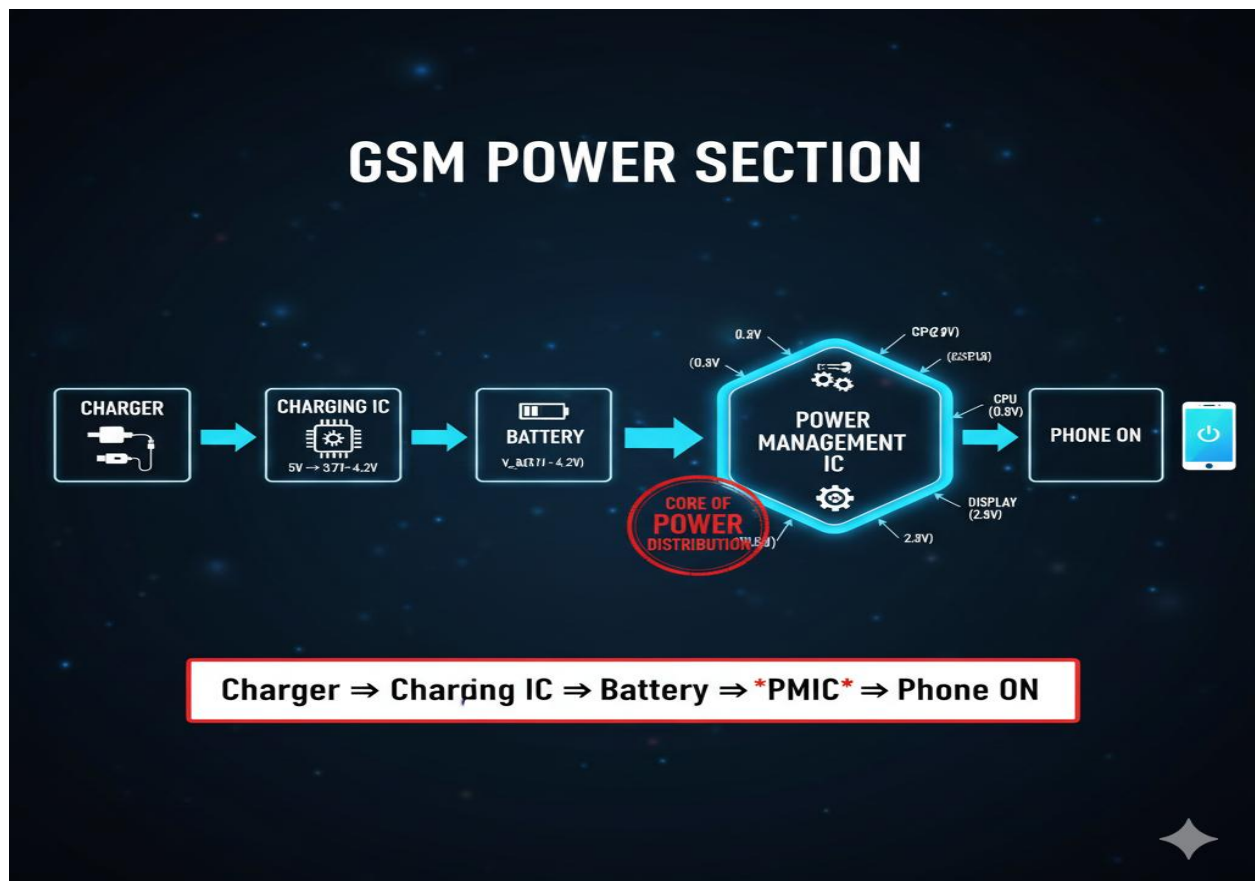
1. Watch a video of a "board-level" phone repair (search for "iPhone dead short repair").
2. Write down three tools the technician used that we did *not* use for basic PC repair.
3. Write one paragraph describing the *fault* and the *solution* you observed.



WEEK 2: GSM POWER SECTION

2.1 INTRODUCTION

The "Power Section" is the most critical part of the phone. It's the "heart and circulatory system" that manages all power, from the charger to the battery and to every other component. A fault here means a dead phone.



2.2 THE PATH OF POWER (SIMPLIFIED)

1. **Charger** -> **Charging Port** -> (This is the 5V-12V from your charger, called **VBUS**)
2. -> **Over-Voltage Protection (OVP) IC** ->
3. -> **Charging IC** ->
4. -> **Battery Connector (VBATT)** ->
5. -> **Battery** ->
6. -> **PMIC (Power Management IC)** ->

7. -> All other ICs (CPU, Screen, etc.)

2.3 KEY COMPONENTS OF THE POWER SECTION

1. BATTERY CONNECTOR

- **What it is:** The physical socket on the motherboard where the battery flex cable plugs in.
- **How to Test:**
 - **Voltage Test (Battery IN):** Set DMM to DC Volts. Place probes on the connector's positive (VBATT) and negative (GND) terminals. You should read the battery's voltage (e.g., 3.7V - 4.2V).
 - **Continuity Test (Battery OUT):** Set DMM to Continuity. Check that the GND pin beeps to ground. Check that the VBATT pin does *not* beep to ground (if it does, you have a dead short).
- **Common Faults:**
 - **Corrosion:** Green/white powder from liquid damage. Must be cleaned (with Isopropyl Alcohol).
 - **Cold Solder Joint:** The pins are "cracked" from the board due to impact. Needs to be re-soldered.

2. CHARGING IC (OR OVP IC)

- **What it is:** A chip that manages the *charging* of the battery. It takes the 5V+ (VBUS) from the charger and "steps it down" to safely charge the battery.
- **Common Faults & Symptoms:**
 - **"Fake Charging":** The phone shows the charging animation, but the battery percentage *never* goes up (or goes down).
 - **No Charging:** The phone doesn't detect a charger at all.
 - **"Charging Slowly":** The IC is not "fast charging" correctly.
 - **Overheating:** The chip itself gets extremely hot when a charger is plugged in.
- **Repair:** This is an IC that must be replaced using a hot air rework station.

3. PMIC (POWER MANAGEMENT IC)

- **What it is:** The "main boss" of power. This is a large, complex IC that takes the main battery power (VBATT) and distributes *many* different, small, stable voltages to every other part of the phone (e.g., 1.8V for the camera, 3.0V for the display, 0.9V for the CPU).
- **Common Faults & Symptoms:**
 - **Completely Dead Phone:** This is the #1 cause of a dead phone (after a dead battery).
 - **Random Functions Not Working:** (e.g., "Wi-Fi is greyed out," "Flashlight doesn't work") because the PMIC is not providing power to that specific sub-system.
 - **Boot Looping:** The phone tries to turn on, the PMIC fails, and it restarts.
- **Repair:** This is an advanced repair. The chip is often "underfilled" (glued) and must be carefully removed (hot air) and replaced (reballing or a new chip).

EVALUATION (WEEK 2)

1. What is "VBUS," and what voltage is it typically?
2. What is the difference between the "Charging IC" and the "PMIC"?
3. A client says their phone "shows it's charging, but the number never goes up." What component is the most likely suspect?
4. You test a battery connector in continuity mode (with the battery out). The positive pin (VBATT) "beeps" to ground. What does this mean?
5. List the (simplified) path of power from the charging port to the battery.

ASSIGNMENT (WEEK 2)

1. Research the charging process. What is "fast charging" (e.g., Qualcomm Quick Charge, USB-PD)?
2. Write a short paragraph explaining how it is different from standard 5V charging. (Hint: It involves communication between the charger and the Charging IC).

WEEK 3: GSM AUDIO SECTION

3.1 INTRODUCTION

The audio section is complex because it involves multiple "input" (microphones) and "output" (speakers) components, all controlled by one "brain" (the Audio IC).



3.2 AUDIO OUTPUT COMPONENTS

1. Earpiece Speaker:

- **Function:** The small speaker you hold to your ear during a call. It is low-power and optimized for human voice frequencies.
- **Location:** Top of the phone, behind a small mesh.
- **Common Faults:**
 - **Clogged Mesh:** The #1 fault. Dirt, dust, and makeup block the hole.
Solution: Clean carefully with a brush and Isopropyl Alcohol (IPA).

- **Torn Flex Cable:** Damaged during a bad screen replacement.
- **Blown Speaker:** A "tinny" or rattling sound. Needs replacement.

2. Loudspeaker (Ringer):

- **Function:** The main speaker for music, speakerphone, and ringtones. It is high-power.
- **Location:** Usually at the bottom of the phone.
- **Common Faults:**
 - **Liquid Damage:** Water enters the speaker grill and corrodes the contacts.
 - **Blown Speaker:** Rattling or no sound at high volume.

3.3 AUDIO INPUT COMPONENTS

1. Microphone (Mic):

- **Function:** Captures your voice. Most phones have *at least* two.
- **Primary Mic:** At the bottom, by the charging port (for calls).
- **Secondary/Noise-Cancelling Mic:** At the top or back, used for video recording and to cancel out background noise during calls.
- **Common Faults:**
 - **Clogged Hole:** The tiny hole next to the charging port is blocked with dirt.
Solution: Clean with a thin needle (carefully!).
 - **Faulty Mic:** Needs to be replaced. This is often *soldered* to a small board, requiring a hot air station.

3.4 THE "BRAIN": AUDIO IC (CODEC)

- **What it is:** The "Audio Codec" (Coder-Decoder) is an IC on the main logic board. It takes digital signals from the CPU (e.g., an MP3 file) and *decodes* them into analog signals for the speaker. It also takes analog signals from the mic and *codes* them into digital for the CPU.
- **Symptoms of Failure:**
 - *All* audio functions fail (mic, speaker, earpiece).
 - "Headphones" are stuck on, even when none are plugged in.

- Audio is distorted, robotic, or has static.
- **Famous Example:** The "iPhone 7 Audio IC Disease." A known fault where the chip loses connection, causing a boot loop and no audio.

3.5 LOGICAL TROUBLESHOOTING (AUDIO)

A client says, "People can't hear me."

1. **Test 1 (Software):** Open the "Voice Memos" app and record. Can you hear playback?
 - *If YES:* The mic is working. The problem is the *network* or the *other person's* phone.
 - *If NO:* The mic is faulty.
2. **Test 2 (Isolate the Mic):** Now, open the Camera app and record a video.
 - *If the video has sound:* The *secondary* (video) mic is working, but the *primary* (call) mic is dead.
3. **Test 3 (Hardware):**
 - Visually inspect the mic hole at the bottom. Is it clogged? Clean it.
 - If still no-go, the component (mic or audio IC) has failed.

EVALUATION (WEEK 3)

1. What is the *most common* cause of a low-volume earpiece speaker, and what is the fix?
2. Why do phones have more than one microphone?
3. What is an "Audio IC" (Codec), and what does it do?
4. A client complains of "no sound at all" from the earpiece, loudspeaker, AND microphone.
Is the problem likely with the components or the IC? Why?
5. What is the *first* thing you should do if a client says, "people can't hear me"?

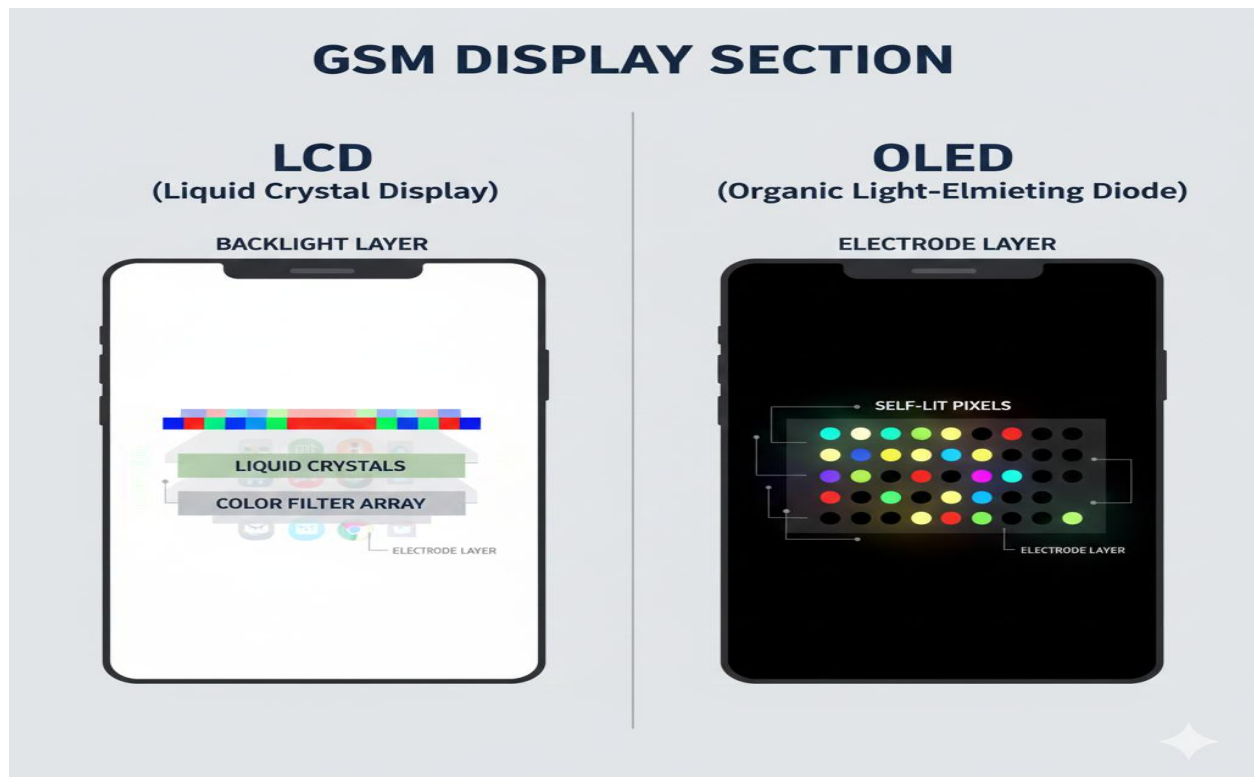
ASSIGNMENT (WEEK 3)

1. Research the "iPhone 7 Audio IC Disease."
2. Write a short summary of its *symptom* (what the user sees) and its *root cause* (what is happening on the board).

WEEK 4: GSM DISPLAY SECTION

4.1 INTRODUCTION

Screen replacement is the "bread and butter" of many repair shops. It is a high-demand, high-profit, but high-risk repair. Understanding the technology is key to avoiding costly mistakes.



4.2 DISPLAY TECHNOLOGIES: LCD vs. OLED

1. LCD (LIQUID CRYSTAL DISPLAY)

- **How it works:** An LCD screen is a "sandwich." It has a layer of liquid crystals, but these crystals *do not produce their own light*. They need a **separate backlight** (a sheet of bright LEDs) to shine *through* them.
- **What this means:** An LCD has *two* main connections: one for the display data and one for the backlight.

- **Common Fault:** "Half-display" or "No backlight." The phone is on, you can hear it, you might even see a *very* faint image under a bright light, but the screen is dark. This is often a *backlight circuit* failure on the board, not a bad screen.
- **Found in:** iPhones (up to iPhone X/11), and most budget/mid-range Android phones.

2. OLED / AMOLED (ORGANIC LIGHT EMITTING DIODE)

- **How it works:** Each individual pixel (sub-pixel) is an "organic" material that *emits its own light* when electricity is applied.
- **What this means:**
 - **No Backlight:** This allows for thinner phones.
 - **Perfect Blacks:** To show "black," the pixel just *turns off*. This gives infinite contrast.
 - **"Always-On Display":** The phone can light up just a few pixels to show the time, using almost no power.
- **Common Fault:**
 - **"Burn-in":** If a static image (like a keyboard or status bar) is left on-screen for too long, it can "stain" the display permanently.
 - More fragile and expensive than LCD.
- **Found in:** Samsung (AMOLED), high-end iPhones (X and newer), and high-end Androids.

4.3 SCREEN REPLACEMENT (OVERVIEW)

This is a delicate process that requires patience and a strict procedure.

- **Tools:**
 - **Heat:** Heat gun or a "heat mat" (set to ~80°C) to soften the adhesive.
 - **Prying:** Suction cup, plastic prying tools (guitars), thin metal "iSesamo" tool.
 - **Screwdrivers:** Precision kit (Pentalobe for iPhone, small Phillips).
 - **Adhesive:** T-7000/B-7000 glue or pre-cut adhesive strips.
- **THE CRITICAL 10-STEP PROCESS (GENERAL):**

1. **Heat** the phone (usually the back for Androids, the front for iPhones) to soften the glue.
2. **Pry** and open the phone *slowly* and *carefully*. Be 100% aware of where the delicate flex cables are.
3. **DISCONNECT THE BATTERY FIRST!** This is the most important step. *Never* work on a phone with the battery connected. You can short the board and destroy it.
4. Disconnect the old screen's flex cables.
5. **TEST THE NEW SCREEN.** Connect the *new* screen (before installing it) and *then* the battery. Power on. Test:
 - Display (any dead spots? correct colours?)
 - Touch (drag an icon all over the screen. Does it follow?)
 - Brightness.
6. **Power off, disconnect battery**, and disconnect the test screen.
7. **Transfer Components:** You must *move* small parts from the *old* broken screen to the *new* one (e.g., earpiece speaker, front-facing camera, proximity sensor).
8. **Clean:** Scrape *all* the old adhesive and broken glass from the phone's frame.
9. **Apply Adhesive:** Apply fresh glue (e.g., T-7000) or a new adhesive strip.
10. **Install** the new screen, reconnect all cables (screen, *then* battery *last*), close the phone, and clamp it to dry.

4.4 COMMON FAULTS

- **No Touch:** You tested a new screen, and the *touch* doesn't work. This can be a *board-level* fault (e.g., a "Touch IC" failure).
- **Ghost Touch:** The screen acts like it's being touched on its own. This is often a sign of a cheap, low-quality replacement screen.
- **Proximity Sensor Failure:** After the repair, the screen doesn't turn off when you put it to your ear. This means the sensor was *not* transferred or aligned correctly.

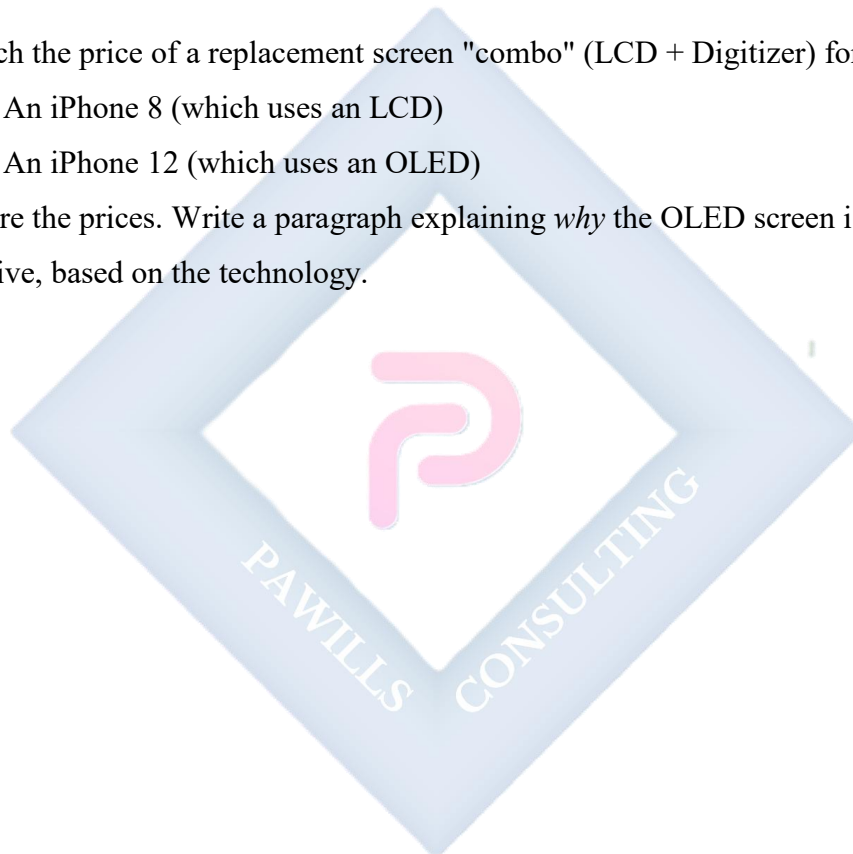
EVALUATION (WEEK 4)

1. What is the main physical difference between an LCD and an OLED display?

2. A client's iPhone 8 screen is black, but you can hear notifications. You shine a bright light on it and see a very faint image. Is the screen broken, or is the *backlight* faulty?
3. What is "screen burn-in," and which technology is susceptible to it?
4. What is the **#1 most important step** to take *after* opening a phone, but *before* disconnecting the screen?
5. Why is it **critical** to test a new screen *before* fully installing it?

ASSIGNMENT (WEEK 4)

1. Research the price of a replacement screen "combo" (LCD + Digitizer) for two phones:
 - An iPhone 8 (which uses an LCD)
 - An iPhone 12 (which uses an OLED)
2. Compare the prices. Write a paragraph explaining *why* the OLED screen is so much more expensive, based on the technology.



WEEK 5: MIDTERM EXAMINATION

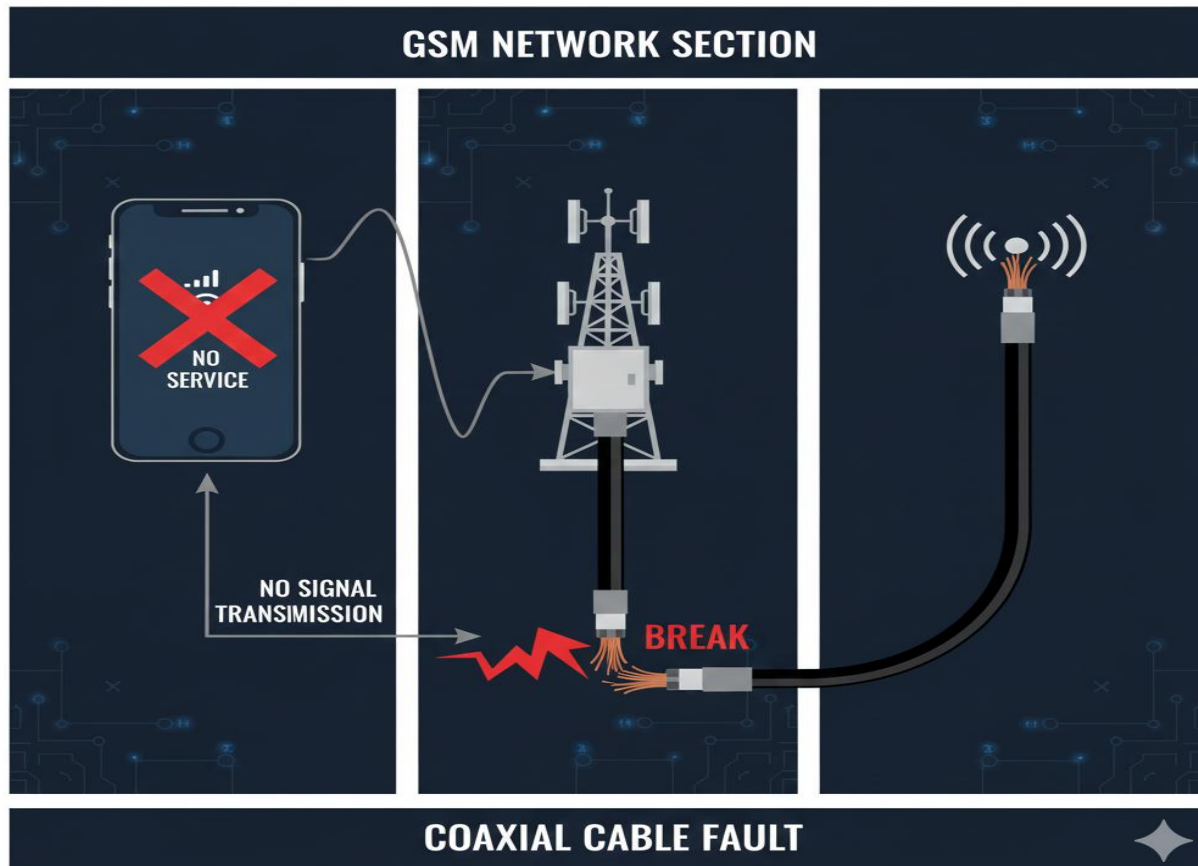
This week is for assessment of all topics covered in Weeks 1-4.



WEEK 6: GSM NETWORK SECTION

6.1 INTRODUCTION

"No Service" is a critical fault. The network section is a complex radio system, but 90% of the faults are caused by simple, physical issues.



6.2 THE LOGICAL "NO SERVICE" TROUBLESHOOTING

A client brings a phone with "No Service."

1. Software/SIM Check (1 min):

- *Airplane Mode*: Is it on? (Turn it off).
- *SIM Card*: Remove the SIM. Is it dirty? (Clean it). Put it in *another* phone. Does it work? (If yes, the SIM is good). Try a *known good* SIM in the client's phone.

- *Check IMEI:* Dial ***#06#**. Does an IMEI number appear?
 - **YES:** Good. The phone's identity is intact.
 - **NO ("Null IMEI" or "0000"):** This is a *software* or *board* (CPU/Baseband) fault. The "radio" part of the phone is not even turning on. No hardware fix will help.
- *Check Blacklist:* Is the phone stolen? (An IMEI can be blacklisted).

2. Physical Hardware Check (The 90% Fault):

- **Antenna System:**
 - The "antenna" is *not* the chip. It's a set of wires, a plastic "antenna box" (often part of the loudspeaker), and a long, thin **coaxial cable** that connects this antenna (at the bottom/top) to the main logic board.
 - **THE MOST COMMON FAULT:** A *previous, bad repair*. A technician replaced a screen or charging port and *forgot* to reconnect the coaxial cable, or they *pinched/tore* the cable.
 - **Solution:** Open the phone. Visually inspect the antenna cable. Is it plugged in at both ends? Is it torn? Replace the cable.

3. Board-Level Hardware Check (The 10% Fault):

- **RF IC (Radio Frequency Transceiver):**
 - **What:** The "radio" chip. It "tunes" to the correct frequency (e.g., 900MHz for MTN).
- **PA (Power Amplifier):**
 - **What:** The "megaphone" chip. It takes the weak signal from the RF IC and *amplifies* it so it can reach the cell tower.
- **Symptoms:** The phone gets very hot in the network section. The IMEI is present, the antenna is connected, but still no signal.
- **Repair:** These are complex BGA (Ball Grid Array) chips that must be identified and replaced.

EVALUATION (WEEK 6)

1. What is the *first* code you should dial on a phone with "No Service," and what are you checking for?

2. What is the *most common* physical cause of a "No Service" fault, especially after a screen repair?
3. What is the difference in function between the "RF IC" and the "PA (Power Amplifier)"?
4. If a phone shows "Null IMEI," is the problem more likely the antenna or the board/software?
5. What is a coaxial cable used for in a phone?

ASSIGNMENT (WEEK 6)

1. Find a "teardown" video (e.g., on iFixit or YouTube) of any modern smartphone.
2. Watch the technician take it apart. Take a screenshot (or sketch) of the **antenna coaxial cable**.
3. Note its path. Where does it start, and where does it end?



WEEK 7: MIDTERM BREAK

This is a break from instruction. Students are encouraged to review the first half of the term's work.



WEEK 8: SOFTWARE FLASHING

8.1 INTRODUCTION

"Flashing" is the common term for re-installing a phone's **firmware** (its Operating System). This is the "software-only" repair that can fix a huge number of problems.



8.2 WHY FLASH A PHONE?

- **Boot Loop:** The phone is stuck on the manufacturer's logo (e.g., "Samsung," "TECNO") and never fully starts.
- **Hard Brick:** The phone appears completely dead *after* a software update failed.
- **Virus/Malware:** The phone is unusable due to severe malware.
- **System Lag:** The phone is extremely slow, and a "Factory Reset" did not help.
- **Forget Password/Pattern:** Flashing will wipe the phone (including the lock). *Note: This will not remove the Google FRP lock (see Week 9).*

8.3 KEY TERMINOLOGY

- **Firmware / ROM:** The OS file package you download. You *must* use the *exact* correct firmware for your phone's model *and* region.
- **Drivers:** Small software pieces for your PC that allow it to recognize and "talk" to the phone in its special modes (e.g., ADB, Fastboot, Qualcomm 9008, MTK VCOM drivers).
- **Flash Tool:** The PC software that "pushes" the firmware to the phone.
- **Flash Mode:** The special state the phone must be in.
 - **Samsung:** "Download Mode" (Volume Down + Home + Power).
 - **Android (General):** "Fastboot Mode" / "Bootloader Mode" (Volume Down + Power).
 - **MediaTek (TECNO/Infinix):** "Preloader Mode" (Phone is OFF. You just plug it in while holding a key).

8.4 COMMON FLASHING TOOLS

- **Odin:** The official (leaked) tool for **Samsung** devices.
- **SP Flash Tool (SmartPhone Flash Tool):** The primary tool for all phones with **MediaTek (MTK)** chips (e.g., most TECNO, Infinix, and many budget phones).
- **Miracle Box / NCK Box / UMT:** Professional "dongles" or "boxes." These are hardware tools that have powerful, all-in-one software for flashing, unlocking, and repairing many different brands (especially Chinese brands).

8.5 THE DANGERS OF FLASHING

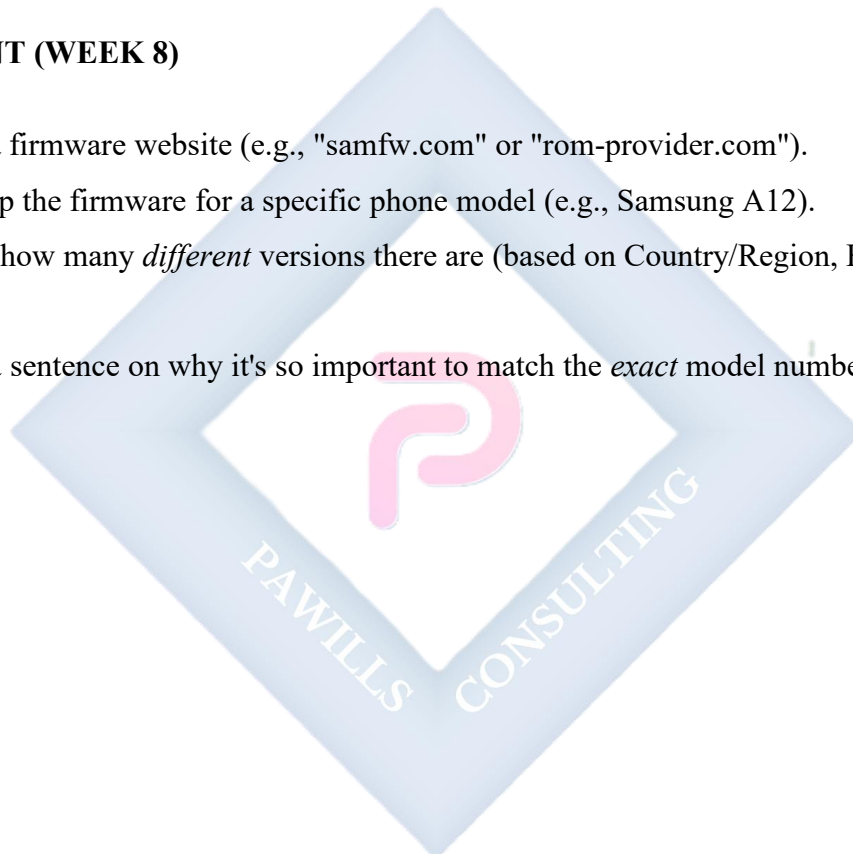
- **Using the Wrong Firmware:** This is the #1 danger. If you flash a "TECNO Spark 5 Pro (KX7)" file onto a "TECNO Spark 5 (KX6)," you can "**hard brick**" the phone. It will be completely dead, and only a very advanced (JTAG or "Test Point") repair can save it.
- **Power Loss:** If the power goes out, or the USB cable disconnects *during* the flash, the phone will be bricked.
- **Data Loss:** Flashing **erases all user data** (photos, contacts, apps). You must warn the client.

EVALUATION (WEEK 8)

1. What is "flashing" a phone?
2. List three common problems that flashing can fix.
3. What is a "boot loop"?
4. What is the *single biggest risk* of flashing, and how do you avoid it?
5. What is the name of the flash tool used for Samsung? What about for MediaTek (TECNO/Infinix)?

ASSIGNMENT (WEEK 8)

1. Go to a firmware website (e.g., "samfw.com" or "rom-provider.com").
2. Look up the firmware for a specific phone model (e.g., Samsung A12).
3. Notice how many *different* versions there are (based on Country/Region, Binary version, etc.).
4. Write a sentence on why it's so important to match the *exact* model number.



WEEK 9: UNLOCKING GSM DEVICES

9.1 INTRODUCTION

"Unlocking" is a broad term. It can mean removing the SIM lock, or it can mean bypassing a security lock. This requires specialized software tools. **Ethical Warning:** These techniques are for *legitimate* repair (e.g., a client forgot their password, or they want to use their phone on a new network). Using them for stolen devices is illegal.



9.2 SIM (NETWORK) UNLOCKING

- **What it is:** Removing the "carrier lock" (e.g., a GLO phone) so that it can be used with any SIM card (e.g., MTN, Airtel).
- **How it's done:**
 1. **Unlock Code:** The "official" way. You ask the original carrier for a code.

2. **Software "Boxes":** Tools like **Z3X Box** (for Samsung) or **Octoplus Box** can "read" the lock from the phone or "patch" the modem to bypass it. This is a direct software operation.

9.3 IMEI REPAIR

- **What it is:** The IMEI (International Mobile Equipment Identity) is the phone's 15-digit "fingerprint." It is unique and identifies your phone to the network.
- **The Fault:** Software corruption (e.g., after a bad flash) can wipe the IMEI, leading to "Null IMEI" or "0000000...".
- **The Symptom: NO SERVICE.** A phone *cannot* register on a network without a valid IMEI.
- **The Repair:**
 - This is *not* changing the IMEI. It is **repairing** or **restoring** the *original* IMEI (which is printed on a sticker on the back of the phone or on the box).
 - **How:** This requires a professional box (like UMT, Miracle Box). You put the phone in a special mode, connect to the tool, and use the "Repair IMEI" function to write the *original* number back to the phone's memory.
- **WARNING:** Changing an IMEI to a *different* number (e.g., from a stolen phone) is a serious crime. We only *repair* a corrupt IMEI to its original, legal number.

9.4 FRP (FACTORY RESET PROTECTION) BYPASS

- **What it is:** This is the "Google Lock." It's an anti-theft feature. If you *hard reset* an Android phone (from recovery mode), when it reboots, it will demand the **"last Google account and password used on this device."**
- **The Problem:** Clients (or 2nd-hand phone buyers) reset their phone and have *no idea* what the old account was. The phone is now a "paperweight."
- **The "Bypass":**
 - This is a constant "cat-and-mouse" game between Google and technicians.
 - Methods change *weekly*.
 - **Examples:**

- **Exploits:** Tricking the phone by using "TalkBack" mode, emergency dialer codes (*#0*#), or connecting to a PC with a "FRP Bypass Tool" (e.t., SamFw Tool).
- **Test Point:** For some (especially MTK) phones, you must open the phone, short two small "test points" on the motherboard, and plug it into the PC. This forces the phone into a special mode that allows the FRP to be wiped.

EVALUATION (WEEK 9)

1. What is the difference between "SIM Unlocking" and "FRP Bypass"?
2. What is an IMEI number, and what *symptom* does a "Null IMEI" cause?
3. Is it legal to *change* a phone's IMEI? Is it legal to *repair* a corrupt IMEI to its original number?
4. What is "FRP (Factory Reset Protection)," and when is it activated?
5. What is a "software box" (e.g., Z3X, UMT)?

ASSIGNMENT (WEEK 9)

1. On your (or a friend's) Android phone, go to the Emergency Dialer.
2. Type in the code *#0*#
3. What happens? (This is a "Test Menu" that is often used as an "exploit" to bypass FRP on Samsung phones). Describe what you see on the screen.

WEEK 10: DATA RECOVERY TECHNIQUES

10.1 INTRODUCTION

"Can you get my photos?" This is the most *emotional* and *difficult* question a technician faces. It is critical to **manage the client's expectations** honestly. In 90% of cases on modern phones, the answer is **NO**.



10.2 THE EASY CASE: MEMORY CARD (SD CARD)

- **The Scenario:** The client has a *removable* SD card. The card is *corrupted*, or they *accidentally* deleted photos.
- **The Solution:**
 1. Put the SD card in a card reader and connect it to a PC.
 2. Use PC data recovery software (e.g., **Recuva** (free), **EaseUS Data Recovery Wizard**, **Disk Drill**).
 3. Scan the card. These tools can find files that have been "deleted" (but not yet overwritten).
- **The Catch:** If the card is *physically* broken or "not detected" at all, this won't work.

10.3 THE MODERN NIGHTMARE: INTERNAL, ENCRYPTED STORAGE

- **The Problem:** All modern smartphones (iOS and Android 6.0+) use **File-Based Encryption (FBE)**.
- **What this means:** All your data (photos, contacts) is *encrypted* on the storage chip. The "key" to unlock this data is *your* password/PIN/pattern.
- **THE HARD TRUTH:**
 1. **If the phone is dead (e.g., PMIC failure, dead board):** The data is *gone*. It is encrypted on a chip that we cannot access.
 2. **If the client forgot the password:** The data is *gone*. The *only* way to "fix" the phone is a Factory Reset, which *erases* the data (and the key).
 3. **If the phone is liquid-damaged:** The data is *gone*.

10.4 THE ONLY SCENARIOS WHERE RECOVERY IS POSSIBLE:

- **SCENARIO 1: Broken Screen (Phone is ON)**
 - **Problem:** The screen is black or "touch" is broken, but the phone itself *works*.
 - **Solution (USB-OTG):** This is the best trick.
 1. Get a **USB-C Hub** (or adapter) that has an **HDMI port** and a **USB-A port**.
 2. Plug the Hub into the phone.
 3. Plug a **USB Mouse** into the Hub.
 4. Plug an **HDMI cable** into the Hub and a TV/Monitor.
 5. You can now see the phone's screen on the TV and *control it* with the mouse.
 6. Enter the password, go to the gallery, and back up the data (e.g., to Google Photos).
- **SCENARIO 2: "Chip-Off" Recovery (The "Pro" Level)**
 - **Problem:** The phone board is dead, but the *client* is a high-value (e.g., corporate/government) and willing to pay *a lot* (N100,000+).
 - **Solution:** This is extremely advanced.
 1. Use a hot air station to **de-solder** the main **UFS/eMMC** (storage) chip from the board.

2. "Reball" (clean and put new solder balls on) the chip.
 3. Place the chip in a *specialized, expensive chip reader* (e.g., Easy-JTAG).
 4. This *still* only gives you the *encrypted* data. You then need advanced forensic software to try and "brute-force" or decrypt the data (often by reading the key from the CPU).
- **This is a forensic, not a repair, skill.**

EVALUATION (WEEK 10)

1. Why is data recovery on a modern (Android 10) phone so much harder than on an old (Android 4) phone?
2. A client brings a phone with a shattered, black screen. The phone is *on* (it rings when you call it). Describe the *exact* procedure for (possibly) recovering their photos.
3. What is "File-Based Encryption (FBE)"?
4. What is "Chip-Off" recovery?
5. What is the *first* tool you should try if a client just "accidentally deleted" photos from their **SD Card**?

ASSIGNMENT (WEEK 10)

1. Research "USB-OTG Adapter" (On-The-Go).
2. List three *other* things you can do with a USB-OTG adapter on a smartphone (besides data recovery).

WEEK 11: REVISION



This week is for a comprehensive review of all topics covered in the Second Term.

- **GSM Power (Week 2):** The "Path of Power" (VBUS -> Charging IC -> VBATT -> PMIC). Faults like "fake charging" (Charging IC) vs. "dead phone" (PMIC/Short).
- **GSM Audio (Week 3):** Components (Earpiece, Loudspeaker, Mic). Faults like clogged mesh/hole. The "brain" (Audio IC) controls all.
- **GSM Display (Week 4):** LCD (needs backlight) vs. OLED (self-lit). The critical 10-step replacement process (DISCONNECT BATTERY FIRST, TEST NEW SCREEN).
- **GSM Network (Week 6):** Logical check (SIM, IMEI *#06#). Physical check (antenna coaxial cable). Board-level (RF IC, PA).
- **Software Flashing (Week 8):** Why? (Boot loop, virus). Tools (Odin, SP Flash Tool). Risk (Hard brick from wrong firmware).
- **Unlocking (Week 9):** SIM Lock (carrier), IMEI Repair (restoring *original*), FRP Bypass (Google lock).

- **Data Recovery (Week 10):** Manage expectations. Modern encryption (FBE) makes it nearly impossible *unless* the phone is on and the *only* fault is a broken screen (use USB-OTG + HDMI + Mouse).



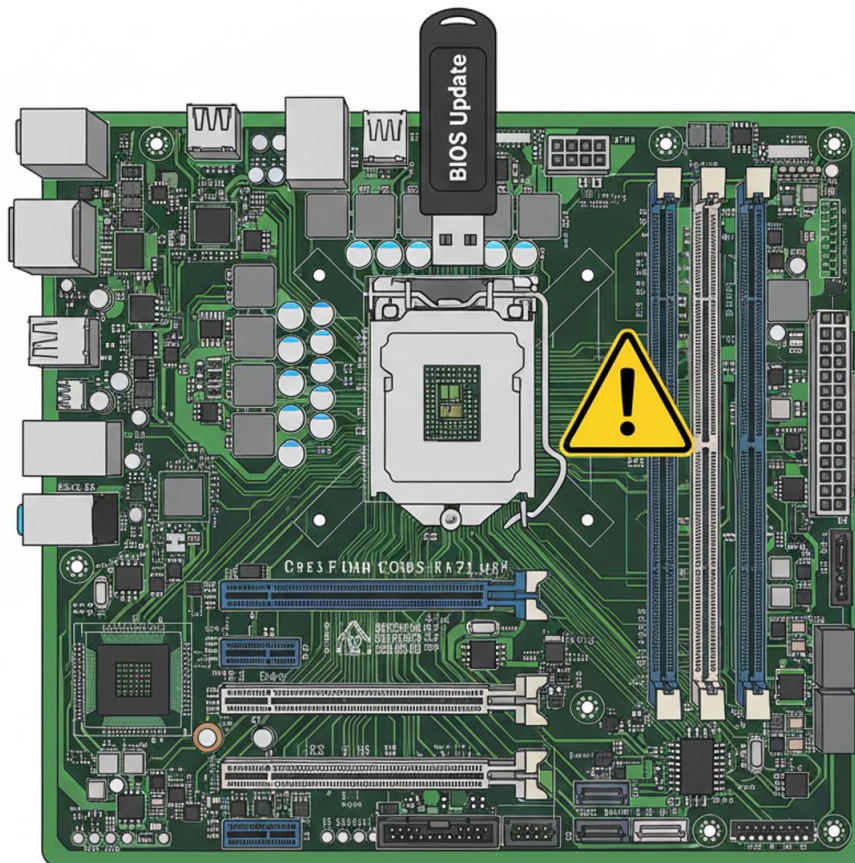
SS2 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

THIRD TERM

WEEK 1: ADVANCED COMPUTER REPAIRS I

1.1 INTRODUCTION

Welcome to the Third Term. We will now combine all our hardware and software knowledge to perform advanced, high-level repair and maintenance tasks. This week focuses on the core software of the motherboard (the BIOS) and the core software of the PC (the Operating System).



1.2 THE BIOS/UEFI: UPDATE AND MANAGEMENT

A. WHAT IS THE BIOS/UEFI?

- **BIOS (Basic Input/Output System):** The "old" firmware. It's the first software that runs when you power on a PC. It initializes and tests the hardware (the POST), and then "boots" the operating system.
- **UEFI (Unified Extensible Firmware Interface):** The modern replacement for BIOS. It does the same job but is much more advanced:
 - It has a graphical interface (you can use a mouse).
 - It supports large hard drives (> 2TB).
 - It has "Secure Boot" (a security feature to prevent malware from loading *before* the OS).

B. WHY UPDATE THE BIOS/UEFI?

A BIOS update is *not* a regular maintenance task. It is done for specific reasons. "**If it ain't broke, don't fix it**" is a good rule here.

1. **Hardware Compatibility:** The *most common* reason. You buy a new CPU (e.g., a new Ryzen 5000) but your motherboard is an older model. The PC won't boot. You must update the BIOS (using your *old* CPU) so it "learns" how to talk to the *new* one.
2. **Security Patches:** To fix critical vulnerabilities (like "Spectre" or "Meltdown") that could allow hackers to attack the firmware.
3. **New Features:** The manufacturer may add new options, like better fan control or improved system stability.

C. THE DANGERS OF A BIOS UPDATE

This is the single most *dangerous* software operation you can perform on a PC.

- **RISK: "Bricking" the Motherboard.** If the update process fails, the PC loses power, or you use the *wrong* BIOS file, the motherboard's firmware chip will be corrupted. The motherboard will be "bricked" — a 100% dead, electronic "brick." It will not power on at all, and it is *extremely* difficult to recover from.

D. THE PROCESS (HOW TO SAFELY UPDATE):

1. **Identify:** Get the *exact* model and revision number of the motherboard (e.g., "GIGABYTE B550 AORUS ELITE rev 1.0").
2. **Download:** Go *only* to the manufacturer's official website. Go to "Support" -> "Downloads" for that exact model. Download the latest BIOS file.
3. **Prepare:** Unzip the file and copy it to a blank, FAT32-formatted USB flash drive.
4. **Update (The Safe Way):**
 - Reboot the PC and enter the BIOS setup (press DEL, F2, or F10).
 - Find the built-in update utility (e.g., "Q-Flash," "EZ Flash," "M-Flash").
 - Select the file from the USB drive and begin the update.
 - **DO NOT TOUCH THE PC.** Do not turn it off. Let it run (it will take 2-5 minutes). It will restart on its own.
5. **Verify:** Enter the BIOS again and check that the new version number is displayed.

1.3 OPERATING SYSTEM (OS) REINSTALLATION

A. WHY REINSTALL THE OS?

1. **Severe Virus/Malware:** The PC is so infected that an antivirus cannot clean it.
2. **System Corruption:** Key Windows files are damaged ("BOOTMGR is missing," random BSODs) and sfc /scannow fails.
3. **Extreme Slowness:** The PC is slow, and a "reset" or "refresh" didn't work.
4. **New Hard Drive:** You are upgrading from an HDD to a new, blank SSD.

B. THE PROCESS: A "CLEAN INSTALL"

A "Clean Install" means *all* data on the main drive will be erased.

1. **BACK UP:** You *must* back up all the client's data (Documents, Photos, etc.) to an external hard drive *first*.
2. **Create Bootable Media:** On a separate, working PC, use the "Windows Media Creation Tool" (from Microsoft's website) to create a bootable Windows 10/11 USB flash drive.
3. **Boot from Media:**
 - Plug the USB drive into the "broken" PC.

- Power on and enter the **Boot Menu** (press F12, F11, or F8).
 - Select the USB drive as the boot device.
4. **Run the Installer:**
- The Windows Setup will load.
 - Enter the product key (or click "I don't have one").
 - Choose "**Custom: Install Windows only (advanced)**".
5. **Partitioning (The Critical Step):**
- You will see a list of partitions on the drive.
 - For a true "clean install," you must **DELETE** all the old partitions (e.g., "System Reserved," "Recovery," "Primary") *until you have one single "Unallocated Space."*
 - Select that "Unallocated Space" and click **Next**.
6. **Installation:** Windows will now handle everything. It will copy files, create new partitions, and install the OS.
7. **Post-Install:**
- **Drivers:** Windows will get most drivers, but you *must* go to the PC/motherboard manufacturer's website and download the specific drivers for the **Chipset, Audio, and GPU**.
 - **Updates:** Run Windows Update until it is 100% up-to-date.
 - **Restore:** Copy the client's data back from the external drive.

EVALUATION (WEEK 1)

1. What is the main difference between the "old" BIOS and the "new" UEFI?
2. What is the *most common* reason to perform a BIOS update?
3. What is "bricking" a motherboard, and how does it happen?
4. What is the difference between clicking "Upgrade" and "Custom" during Windows setup? Which one is a "Clean Install"?
5. What are the *first* and *last* steps of any OS reinstallation? (Hint: Data...)

ASSIGNMENT (WEEK 1)

1. Find the motherboard model of your school or home computer.
2. Go to the manufacturer's website (e.g., Dell, HP, Gigabyte) and find the "Downloads" page for that model.
3. Write down:
 - The latest BIOS version number and its release date.
 - What the update "Fixes" or "Improves" (this is in the description).
 - The latest "Chipset Driver" version.



WEEK 2: ADVANCED COMPUTER REPAIRS II

2.1 INTRODUCTION

Upgrading hardware is the best way to improve a PC's performance. As a technician, you must know how to choose the *correct* parts and how to install them. Compatibility is *everything*.



2.2 HARDWARE UPGRADES: RAM

- **Why Upgrade?** To improve multitasking. The PC can run more programs (e.g., Chrome, Word, Spotify) at the same time without "hanging" or "swapping" to the disk.
- **Compatibility Check (Crucial):**
 1. **Form Factor:** Is it a Desktop or a Laptop?
 - **Desktop:** Uses **DIMM** (Dual In-line Memory Module).
 - **Laptop:** Uses **SO-DIMM** (Small Outline DIMM) - .
 2. **Type (DDR Version):** The motherboard *only* supports one type. You *cannot* put DDR4 RAM in a DDR3 slot.
 - DDR3 (Old), DDR4 (Common), DDR5 (Newest).
 3. **Speed (MHz):** The "speed" of the RAM (e.g., 3200MHz). You should match the speed of the RAM already in the system.

4. **Capacity:** The motherboard has a *maximum* capacity (e.g., "Max 64GB").
- **Installation (Desktop):**
 1. Power off, unplug, use an anti-static strap.
 2. Open the plastic "clips" at both ends of the RAM slot.
 3. Align the **notch** on the RAM stick with the "key" in the slot. (You cannot put it in backwards).
 4. Press down *firmly* on both ends of the stick until the clips "click" into place and lock the RAM.
 5. Boot the PC. It may "memory train" (restart a few times). Check the BIOS or Task Manager to confirm the new amount is seen.

2.3 HARDWARE UPGRADES: HDD TO SSD

- **Why Upgrade?** This is the **single best upgrade** to make an old PC "feel new." It improves boot time, application loading, and overall system "snappiness" by 10x-20x.
- **The Process:** You have two choices:
 1. **Clean Install (Recommended):** Follow the "OS Reinstallation" steps from Week 1. Install the new SSD, boot from the USB, and install Windows fresh onto the new drive.
 2. **Cloning (Advanced):**
 - **What:** "Cloning" copies an *exact* 1:1 image of your old HDD (including Windows, all files, all programs) onto the new SSD.
 - **How:** Connect *both* drives to the PC. Use "cloning software" (e.g., Macrium Reflect, Clonezilla). Run the software to copy "Source" (HDD) to "Destination" (SSD).
 - **Pro:** The user keeps all their data and programs, exactly as they left them.
 - **Con:** It can be complex, and if the old install was corrupt, you've just copied the corruption.

2.4 HARDWARE UPGRADES: GRAPHICS CARD (GPU)

- **Why Upgrade?** For gaming, 3D modelling, or video editing. To get higher "frames per second" (FPS) or render videos faster.
- **Compatibility Check (The Most Complex):**
 1. **Slot:** Does the motherboard have a **PCIe x16** slot? (All modern ones do).
 2. **Physical Size:** This is a *major* issue. A high-end card is *long* and *thick*. Will it physically *fit* inside the PC case? You must measure.
 3. **PSU (Power Supply Unit) (CRITICAL):**
 - A new GPU is *very* power-hungry.
 - Check the "Recommended PSU Wattage" for the card (e.g., a new GPU might need a 650W PSU).
 - Check if your PSU *has* the correct power connectors (e.g., one or two 8-pin PCIe power cables).
 - **If your PSU is too weak, the PC will *not* boot or will crash under load.**
- **Installation:**
 1. Power off, unplug.
 2. Remove the metal "PCIe brackets" from the back of the case.
 3. Align the card with the PCIe x16 slot and press it firmly into place.
 4. Secure the card to the case with screws.
 5. Plug in the required PCIe power cables from the PSU.
 6. Boot the PC. Go to the NVIDIA or AMD website and install the *latest drivers*.

EVALUATION (WEEK 2)

1. What is the physical difference between desktop RAM (DIMM) and laptop RAM (SO-DIMM)?
2. What is the *single best* upgrade you can perform to make a slow, old laptop feel fast?
3. What are the two main methods for moving from an old HDD to a new SSD?
4. What are the *three* (3) main compatibility checks you *must* perform before recommending a new GPU?
5. What happens if you install a powerful new GPU but your PSU is too weak?

ASSIGNMENT (WEEK 2)

1. A client has a "Dell Optiplex 7010" desktop. They want to upgrade its RAM.
2. Research that model. Write down:
 - What *type* of RAM does it use (e.g., DDR3)?
 - What is the *maximum* amount of RAM it supports?
 - Is its RAM DIMM or SO-DIMM?



WEEK 3: COMPUTER VIRUSES AND SECURITY

3.1 INTRODUCTION

As a technician, you are a "PC doctor." You must be able to identify, remove, and (most importantly) *prevent* infections. "Malware" (Malicious Software) is the general term for any program designed to cause harm.



3.2 TYPES OF MALWARE (KNOW YOUR ENEMY)

- **Virus:** The "classic" malware. It "infects" a program by attaching its code to it (e.g., to winword.exe). It needs a *human* to run that program to spread.
- **Worm:** More dangerous. A worm is a *self-replicating* program that spreads across a network *on its own*. It actively looks for other vulnerable PCs to infect.
- **Trojan (or Trojan Horse):** A program that *disguises* itself as something useful.
 - **Example:** A "free game installer" or a "video player" that, when you run it, *also* secretly installs a backdoor for a hacker.

- **Ransomware:** The most *destructive* malware. It *encrypts* all your personal files (photos, documents) and demands a "ransom" (payment, usually in Bitcoin) to give you the decryption key.
- **Spyware:** A program that secretly records your activity.
 - **Example:** A "keylogger" that records *everything* you type, including your bank passwords and private messages.
- **Adware:** The most common and least dangerous. It just spams you with pop-up ads and unwanted browser toolbars.

3.3 PREVENTION (THE 90% SOLUTION)

The *best* way to "fix" a virus is to *never get it*.

1. **Use Legitimate Software:** The #1 source of Trojans and Ransomware in Nigeria is **cracked software** ("KMS activators," "free Photoshop," "cracked games"). Do not use them.
2. **Keep Everything Updated:**
 - **Windows Update:** Keep it on and updated.
 - **Antivirus:** Keep its "definitions" (the virus database) updated.
 - **Programs:** Update your browser (Chrome, Firefox), as hackers attack vulnerabilities in old software.
3. **Be Smart (Phishing):**
 - Phishing is an *email* or *text message* that *tricks* you.
 - **Example:** "Your bank account has been locked. Click *here* to log in." The link goes to a *fake* website.
 - **Rule:** Never click links in emails. Go to the official website yourself. Be suspicious of *everything*.
4. **Use a Good Antivirus:** Windows Defender (built into Windows 10/11) is now very good and is all most users need.

3.4 REMOVAL (THE 10% SOLUTION)

A client brings in an infected PC.

1. **Isolate:** Disconnect the PC from the internet (unplug the LAN, disconnect Wi-Fi) to stop it from "phoning home" or spreading.
2. **Identify:** Install and run a reputable scanner. A good *second opinion* scanner is **Malwarebytes**.
3. **Scan (In Safe Mode):**
 - Reboot the PC into **Safe Mode**.
 - Why? Safe Mode loads *only* the essential Windows files. The virus (which is a non-essential program) will not be "active" and cannot hide or defend itself.
 - Run the full scan from Malwarebytes.
4. **Quarantine/Remove:** Let the antivirus tool move all found infections to "Quarantine" or delete them.
5. **Clean Up:** Go to "Add or Remove Programs" and uninstall any "junk" (toolbars, "PC Optimizers") that were installed. Reset all web browsers.
6. **If all else fails...**
 - A "Clean Install" (Week 1) is the *only* 100% guaranteed way to remove a deep infection.

EVALUATION (WEEK 3)

1. What is the main difference between a "Virus" and a "Worm"?
2. What is a "Trojan"? Give a common example.
3. A client's files are all renamed ".locked" and they have a text file demanding Bitcoin. What type of malware is this?
4. What is the *single biggest* source of malware infections from user behavior?
5. Why is it better to run an antivirus scan in "Safe Mode"?

ASSIGNMENT (WEEK 3)

1. Research the term "Phishing."
2. Find one example of a real phishing email (you can search for "phishing email examples").

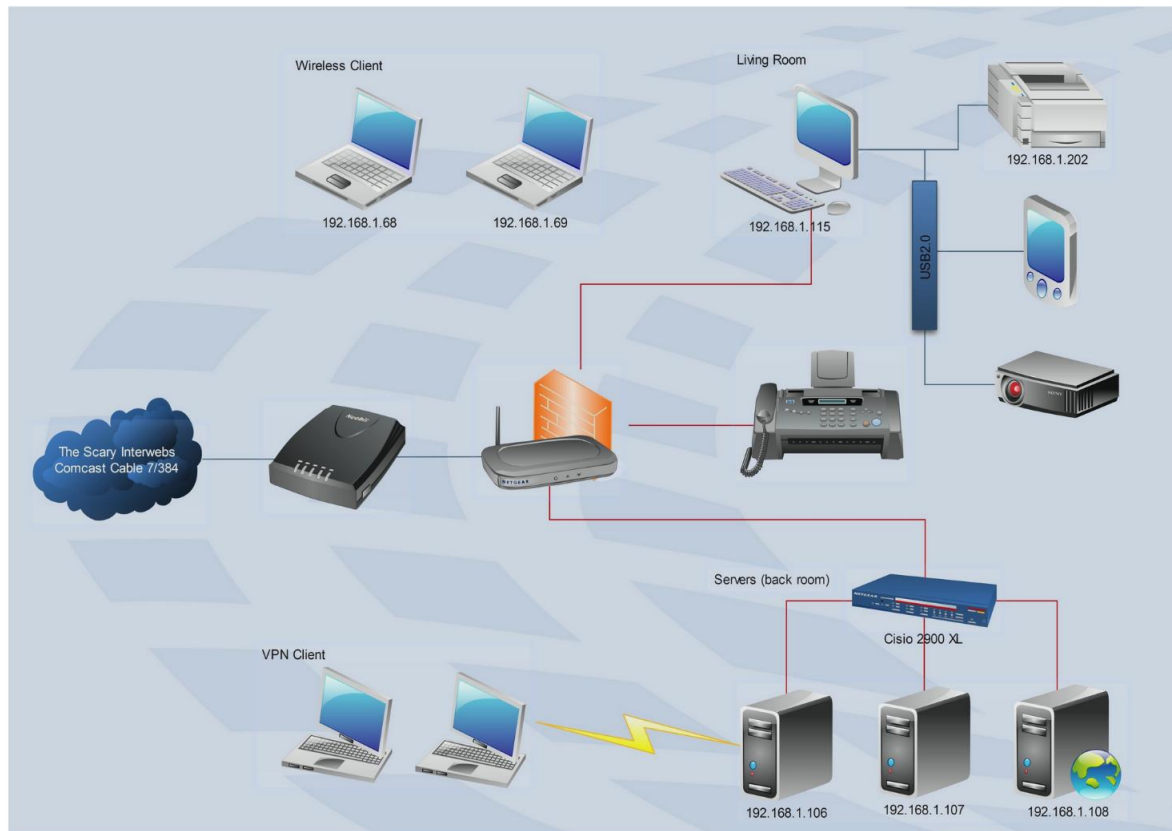
3. List 3 "red flags" (clues) that prove the email is fake. (e.g., "The email address is from @gmail.com, not @gtbank.com," "There are spelling mistakes," "The link (hovering over it) goes to a strange website").



WEEK 4: NETWORKING SETUP

4.1 INTRODUCTION

No PC is an island. Connecting computers to each other and to the internet is a core task. This is "Networking."



4.2 LAN (LOCAL AREA NETWORK)

- **What it is:** A network in a *small, local* area (a home, an office, a school).
- **Key Components:**
 1. **Router:** The "gateway." This device connects your *entire* LAN to the *internet*. It's the "boss."
 2. **Switch:** A "power strip" for the network. It allows you to connect *many* wired devices (PCs, printers) to the router. (Most home routers have a 4-port switch built-in).

3. **Ethernet Cables (Cat 5e/Cat 6):** The wired "plumbing."
 4. **RJ45 Connector:** The "head" of an Ethernet cable.
- **IP Addressing:**
 - Every device on the network needs a unique "address" (an IP address, e.g., 192.168.1.10).
 - **DHCP (Dynamic Host Configuration Protocol):** This is the "automatic" system. The *router* runs a DHCP server that *automatically* assigns an IP address to every device that connects. This is the default for 99% of networks.

4.3 WI-FI (WLAN - WIRELESS LOCAL AREA NETWORK)

- **What it is:** A LAN that uses *radio waves* (Wi-Fi) instead of cables.
- **Key Components:**
 1. **Access Point (AP):** The device that "broadcasts" the Wi-Fi signal. In a home, the *router* is also the Access Point.
 2. **SSID (Service Set Identifier):** The "name" of the Wi-Fi network (e.g., "MTN 5G Hotspot," "Tunde's_House").
 3. **Security:**
 - **WEP (Obsolete):** Very old, insecure.
 - **WPA2:** The long-time standard. Good security.
 - **WPA3:** The new, most secure standard.
 - **The Password:** You *must* secure your Wi-Fi with a strong password.

4.4 BASIC NETWORK TROUBLESHOOTING

- **Symptom:** "Sir, the internet is not working."
- **THE LOGICAL 5-STEP TEST:**
 1. **Check Physicals:**
 - *You:* Is the Wi-Fi "on"? Is the LAN cable *plugged in* (do you see a light)?
 - *Them:* Check the Router. Are the "Internet" or "WAN" lights on? (Are they red?) The #1 fix is "**reboot the router.**"
 2. **Check IP Address (The ipconfig command):**

- Open Command Prompt (cmd).
 - Type ipconfig and press Enter.
 - **Good:** You see an "IPv4 Address" like 192.168.1.10. This means you are *successfully connected to the router*.
 - **Bad:** You see 169.254.x.x. This is a "junk" address. It means you *failed* to get an address from the router's DHCP. The problem is between you and the router (bad cable, bad Wi-Fi).
3. **Check Gateway (Ping the Router):**
- In ipconfig, find your "Default Gateway" (e.g., 192.168.1.1). This is the router's address.
 - Type ping 192.168.1.1.
 - **Good:** You get "Reply from..." This confirms you can talk to the router.
4. **Check Internet (Ping Google's DNS):**
- Type ping 8.8.8.8.
 - **Good:** You get "Reply from..." This confirms you can reach the *internet*.
 - **Bad:** You get "Request timed out." This means the problem is *not* your PC; the problem is the *Router* or the *Internet Service Provider (ISP)* (e.g., MTN, GLO).
5. **Check DNS (Ping Google's Name):**
- Type ping google.com.
 - **Bad:** ping 8.8.8.8 *worked*, but ping google.com *fails*. This is a **DNS (Domain Name System)** problem. Your PC can't "translate" the *name* google.com into its *IP address*.
 - **Fix:** Manually set your DNS to 8.8.8.8 (Google's) or 1.1.1.1 (Cloudflare's).

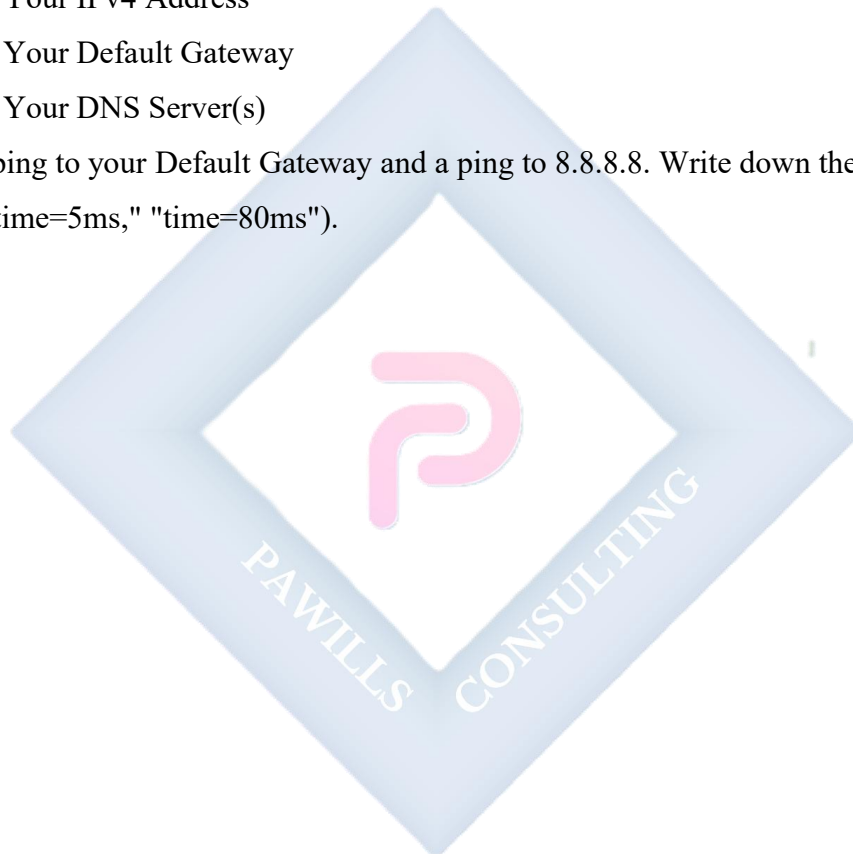
EVALUATION (WEEK 4)

1. What is the difference between a Router and a Switch?
2. What does "DHCP" do, and who (what device) runs it?
3. What is an "SSID"?
4. You type ipconfig and see an address of 169.254.10.20. What does this mean?

5. What is the difference in troubleshooting between ping 8.8.8.8 and ping google.com?
What are you testing for?

ASSIGNMENT (WEEK 4)

1. At home or school, connect to the network.
2. Open cmd and type ipconfig /all.
3. Write down:
 - Your IPv4 Address
 - Your Default Gateway
 - Your DNS Server(s)
4. Run a ping to your Default Gateway and a ping to 8.8.8.8. Write down the "time=XXms" (e.g., "time=5ms," "time=80ms").



WEEK 5: MIDTERM EXAMINATION

This week is for assessment of all topics covered in Weeks 1-4.



WEEK 6: GSM SOFTWARE OPTIMIZATION

6.1 INTRODUCTION

We now move back to GSM, focusing on advanced *software* modifications. Flashing (Term 2) is "repair." This is "optimization" and "customization." These are *powerful* but *dangerous* procedures.



6.2 ROOTING (ANDROID)

- **What it is:** "Rooting" is the process of gaining "**root access**" (or "superuser") permissions on an Android device.
 - By default, you are a "user." You cannot touch the "system" files.
 - "Root" is the "Administrator." You can change or delete *anything*.
- **Why Root? (The "Pros"):**
 1. **Remove Bloatware:** Delete the "junk" apps that your carrier or manufacturer (e.g., TECNO, Samsung) pre-installed and won't let you uninstall.

2. **Advanced Tools:** Run powerful apps that need deep access (e.g., advanced backup tools, firewall tools).
 3. **Custom ROMs:** Rooting is the first step to installing a custom OS.
- **The DANGERS (The "Cons"):**
 0. **voids Warranty:** The second you root, your warranty is gone.
 1. **"Bricking":** If you delete the wrong file (e.g., system.ui), the phone will crash or boot loop.
 2. **Security Risk:** The #1 reason *not* to root. You *and* any malware on your phone *both* have admin access. Banking apps, Google Pay, and Netflix *will refuse to run* on a rooted device.
 - **Modern Rooting (Magisk):**
 - The modern method is "systemless root" using a tool called **Magisk**.
 - It *hides* the root status from other apps, which *may* allow banking apps to work.

6.3 CUSTOM ROMS

- **What is a ROM?** The firmware (OS) is stored in "Read-Only Memory" (ROM). A "Custom ROM" is a modified, unofficial version of the Android OS, built by developers in the community.
- **Why Install One? (The "Pros"):**
 1. **Get Latest Android:** Your phone is 3 years old, and the manufacturer (e.g., Samsung) has *stopped* updating it. A custom ROM (like **LineageOS**) can give you the *latest* Android 13/14 on your old hardware.
 2. **Performance:** They are often "de-bloated" (all junk removed) and run *faster* and *smoother* than the stock firmware.
 3. **Features:** They add new features and customization that the stock OS doesn't have.
- **The Process (Advanced):**
 1. **Unlock the Bootloader:** This is the *first* step. It wipes all your data.
 2. **Install a Custom Recovery:** Flash a tool like **TWRP (Team Win Recovery Project)**. This is a mini-OS that lets you back up and flash new ROMs.

3. **Flash the ROM:** Download the ROM file (a .zip) and the Google Apps (GApps) package.
4. **Wipe & Flash:** Boot into TWRP, *wipe* the System/Data/Cache, and then "Install" the ROM zip.

6.4 OTA (OVER-THE-AIR) UPDATES

- **What it is:** The *official* way to update. The manufacturer "pushes" the update "Over-The-Air" (via Wi-Fi or mobile data).
- **Why Update?**
 - **SECURITY:** 90% of updates are "**Security Patches**" to fix vulnerabilities. This is the most important reason.
 - **Features:** To get new features (e.g., an OS-level update from Android 12 to 13).
- **Technician's Note:** You *cannot* install an OTA update on a *rooted* phone. The update will fail, and it may "soft-brick" the device. You must "un-root" first.

EVALUATION (WEEK 6)

1. What is "rooting"? What does it give you access to?
2. What is the single *biggest* security risk of rooting?
3. What is a "Custom ROM," and what is the #1 reason a user would want one?
4. What is "TWRP," and what is its purpose?
5. What is an "OTA update," and why is it so important to install them?

ASSIGNMENT (WEEK 6)

1. Research "LineageOS."
2. Go to their website ("Supported Devices" list).
3. Find *one* phone model that is supported.
4. Write down the phone model and the *latest* version of Android that LineageOS provides for it. (e.g., "Phone: Samsung Galaxy S5. Latest LineageOS: Android 11").

WEEK 7: MIDTERM BREAK

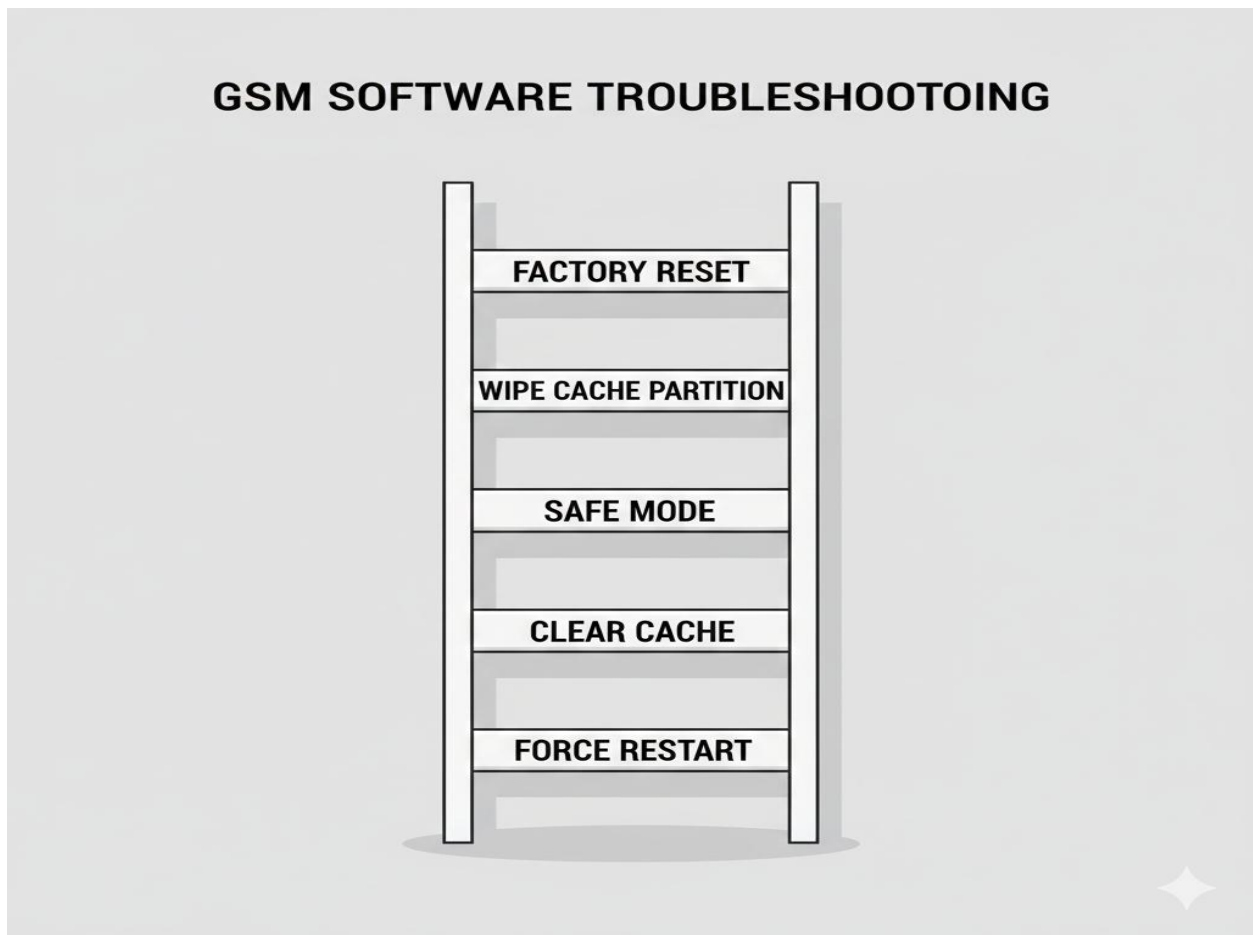
This is a break from instruction. Students are encouraged to review the first half of the term's work.



WEEK 8: TROUBLESHOOTING GSM SOFTWARE ISSUES

8.1 INTRODUCTION

Not all phone problems are hardware. In fact, most (app crashes, slowness, boot loops) are software. As a technician, you must know how to diagnose and fix these.



8.2 THE "SOFTWARE REPAIR" LADDER (LEAST to MOST DESTRUCTIVE)

Always start at Step 1. Do not jump to Step 5 unless you have to.

STEP 1: FORCE RESTART

- **The Problem:** The phone is "frozen" or "stuck."
- **The Fix:** A "Force Restart" (or "Hard Reboot"). This cuts the power *without* deleting data.

- **How:** On most phones, **press and hold the POWER button + VOLUME DOWN button** for 10-15 seconds.
- **Why it works:** It clears the "active RAM" and forces the OS to reload from scratch.

STEP 2: CLEAR APP CACHE/DATA

- **The Problem:** One *specific* app is crashing (e.g., "Instagram keeps stopping").
- **The Fix:**
 1. Go to Settings -> Apps -> [Problem App] -> Storage.
 2. **Clear Cache:** This is the *first* thing to try. It deletes temporary files. No data is lost.
 3. **Clear Data:** This is the *second* thing to try. It resets the app to its "factory" state. (e.g., You will be logged out of Instagram).

STEP 3: BOOT INTO SAFE MODE

- **The Problem:** The phone is crashing, boot looping, or *very* slow. You suspect a *third-party app* (one *you* installed) is the cause.
- **The Fix:** Boot into **Safe Mode**.
- **How:** (Method varies) Usually, press and hold POWER, then *press and hold* the "Power Off" icon on the screen.
- **What it does:** Safe Mode loads *only* the original, "factory" system apps. It *disables* all third-party apps.
- **Diagnosis:**
 - If the phone works perfectly in Safe Mode, you have **100% confirmed** the problem is a third-party app.
 - **Solution:** Reboot normally. Now, uninstall your *most recent* apps one by one until the problem stops.

STEP 4: WIPE CACHE PARTITION

- **The Problem:** The phone is slow, laggy, or boot looping, *especially* after a big "OTA Update."

- **The Fix:** Wipe the **System Cache Partition** from "Recovery Mode."
- **How:**
 1. Power off the phone.
 2. Boot into **Recovery Mode** (e.g., POWER + VOLUME UP).
 3. Use the volume buttons to select "**Wipe Cache Partition.**"
- **Why it works:** This deletes all the "junk" temporary files left over from the old OS. It is **100% safe and deletes NO user data** (no photos, no contacts). This is a *powerful* fix.

STEP 5: FACTORY RESET

- **The Problem:** All the above failed. You have a deep software problem, malware, or you are selling the phone.
- **The Fix:** A **Factory Data Reset**.
- **How:** Settings -> General Management -> Reset -> Factory Data Reset.
- **What it does:** **ERASES EVERYTHING.** All photos, contacts, apps, and accounts are gone. The phone is returned to its "out-of-the-box" software state.
- **WARNING:** This *will not* remove the **FRP (Google) Lock**. You *must* know the Google password to use the phone again (see Term 2, Week 9).

EVALUATION (WEEK 8)

1. What is the difference between "Clear Cache" and "Clear Data" for an app?
2. What is "Safe Mode," and what does it help you diagnose?
3. A client's phone is stuck in a boot loop. What is the *first* thing they should try in Recovery Mode that *won't* delete their photos?
4. When is a "Factory Reset" the correct solution?
5. What is the *most important* warning to give a client before performing a Factory Reset?

ASSIGNMENT (WEEK 8)

1. Find the *exact* key combination to boot into "Recovery Mode" for a (a) Samsung and (b) TECNO/Infinix phone.

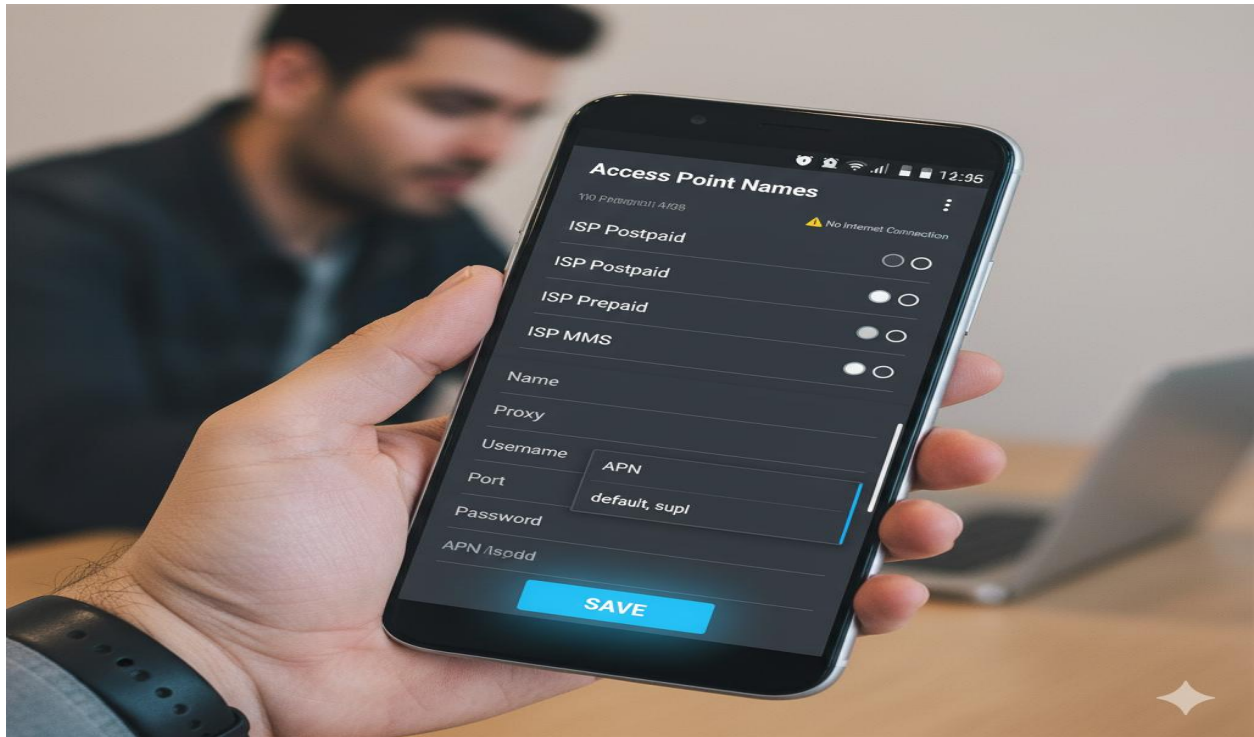
2. Find the *exact* key combination to boot into "Safe Mode" for your own phone (or a popular model).
3. Write these 3 procedures down.



WEEK 9: INTERNET CONNECTIVITY ISSUES

9.1 INTRODUCTION

A phone with no internet is a "dumb" phone. This is a *very* common complaint. The problem can be the carrier, the hardware, or (most often) a *simple software setting*.



9.2 THE LOGICAL TROUBLESHOOTING LADDER (DATA)

- **Symptom:** "My data is not working." (A "!" or "x" is next to the network bars).

STEP 1: THE "CLIENT" PROBLEMS

- "Did you pay your data subscription?" (The #1 problem!)
- "Did you turn on 'Mobile Data' in the settings?"
- "Is 'Airplane Mode' on?"
- **Fix: Restart the phone.** (This re-registers the phone on the network and fixes 50% of problems).

STEP 2: THE "NETWORK" PROBLEMS

- **Symptom:** The data is "on," but *very* slow.
- **Check:** Look at the network icon. Does it say "E" (Edge) or "H+" (HSPA) instead of "4G" (LTE) or "5G"?
- **Fix:**
 - Go to Settings -> Network -> Preferred Network Type.
 - It might be "stuck" on 2G/3G.
 - Set it to "**4G/LTE/3G/2G (Auto)**".
 - This forces the phone to *look* for the faster 4G network.

STEP 3: THE APN (ACCESS POINT NAME) - THE #1 TECHNICIAN FIX

- **What it is:** The APN is a "gateway" setting. It's the *exact* address and set of rules your phone needs to connect to *your carrier's* (e.g., MTN) internet.
- **The Problem:** The APN settings are *missing, corrupt, or wrong*. This often happens when a client brings a phone from another country (e.g., a "UK-used" phone) or after a software update.
- **The Fix (Manual APN Configuration):**
 1. Go to Settings -> Mobile Networks -> Access Point Names (APN).
 2. You will see a list. If it's empty, this is the problem.
 3. Click "Add."
 4. You must now *manually* type in the settings for the carrier.
 - **Example (MTN Nigeria - *this can change*):**
 - **Name:** MTN
 - **APN:** web.gprs.mtnnigeria.net
 - **Username:** web
 - **Password:** web
 - (You must Google "MTN APN settings" or "GLO APN settings" for the current, correct info).
 5. Click **Save**, then *select* the new APN you just made.

6. Turn mobile data off and on. The internet will almost always start working.

STEP 4: HOTSPOT & TETHERING ISSUES

- **Symptom:** "My data works, but my hotspot won't connect."
- **The Fix:** This is *also* often an APN problem.
 1. Go back to the APN settings.
 2. Find the "APN Type" field.
 3. It might just say default.
 4. You must *add* the hotspot setting. Change it to: default,supl,dun
 - dun stands for "Dial-Up Networking," the old term for tethering.
 5. Save and try the hotspot again.

EVALUATION (WEEK 9)

1. A client's data is not working. What are the *first three* "client-level" questions you should ask them?
2. What is the difference between "H+" and "4G/LTE"?
3. What is an "APN," and why is it so important?
4. When would you most likely need to manually configure an APN?
5. If a hotspot is not working, what "type" setting can you add to the APN to fix it?

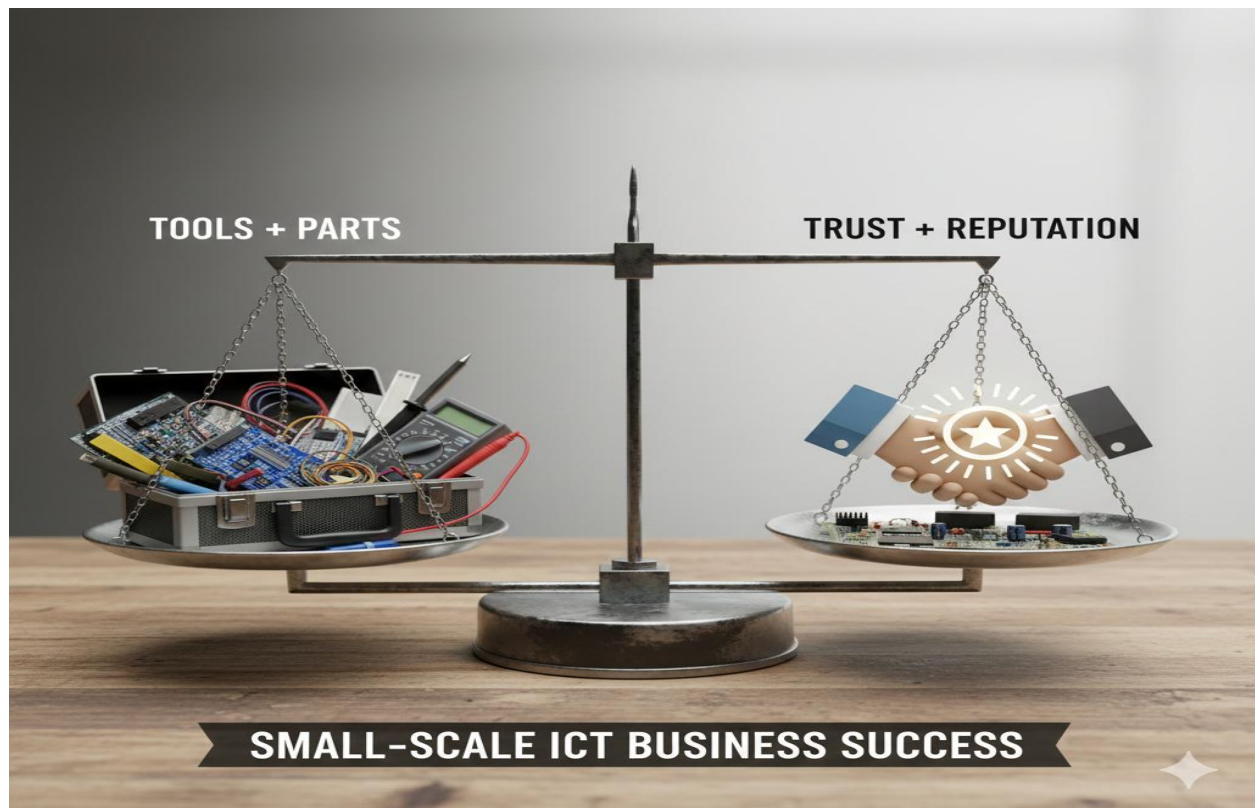
ASSIGNMENT (WEEK 9)

1. Go to Google and find the *current* (2024/2025) APN settings for:
 - GLO
 - Airtel Nigeria
2. Write down the "APN," "Username," and "Password" for both.

WEEK 10: SMALL-SCALE ICT BUSINESS

10.1 INTRODUCTION

This week is the "capstone" of your entire SS2. You have the *technical skills*. But a skill is *not* a business. A business is about **structure, management, and service**.



10.2 THE BUSINESS PLAN

Before you buy your first tool, you need a plan.

- **1. Your Niche (Specialty):** You cannot be "good at everything."
 - *Examples:* "I will be the best iPhone screen repairer in my area." "I will focus on software (flashing, unlocking, viruses)." "I will do custom PC building for gamers." "I will do home/office network setups."
- **2. Your Location:**

- **High-Traffic:** A "kiosk" or small shop near a market or on a busy street. (High cost, high customers).
- **Home-Based:** "I come to you" (mobile repair) or "They come to me" (home office). (Low cost, hard to get customers).
- **3. Your Name:** A registered business name (with the CAC - Corporate Affairs Commission) makes you look professional and allows you to open a company bank account.
- **4. Your Tools & Inventory:**
 - **Tools:** Get a *good* set of tools (e.g., iFixit kit, good DMM, hot air station). Do not buy cheap tools that break or damage devices.
 - **Inventory:** You cannot stock every part. Start by stocking the "Top 5":
 - e.g., Screens for iPhone X, XR, 11
 - e.g., Batteries for Samsung A12, A20
 - e.g., Charging ports for TECNO Spark

10.3 THE "SOFT SKILLS" (MORE IMPORTANT THAN TECHNICAL SKILLS)

- **1. Customer Service (Trust):**
 - **Communication:** Greet people well. *Listen* to their problem. Explain the fault in *simple* terms.
 - **Honesty:** *Never* lie. If you *cannot* fix it, say so. If you *broke* something, *own it* and fix it.
 - **Transparency (Pricing):** Give the client a *clear price* (quote) *before* you start the work. (e.g., "The new screen is N15,000, and my labour (work) is N5,000. The total is N20,000. Do you agree?").
- **2. Data Privacy (The #1 Ethic):**
 - **You are a "data doctor."** You *will* have access to your clients' *private photos, bank alerts, and messages*.
 - Your business will *die* if you get a reputation for being a "gossip" or a "thief."
 - **Rule:** You *never* look. You *never* snoop. You *never* copy a client's files. You *never* share what you saw. You are a professional.
- **3. Marketing:**

- **Google Maps:** Register your business on Google Maps. This is *free* and the #1 way new customers will find you ("computer repair near me").
- **Word-of-Mouth:** Your *best* marketing. A happy client will tell 2 friends. An angry client will tell 20.
- **Social Media:** Use Instagram/TikTok to show *your work*. (e.g., A short video of you "cleaning a dusty PC," or a "before and after" of a screen replacement).

10.4 THE MONEY (BASIC FINANCES)

- **Pricing:** Total Price = Cost of Part + Labour
 - **Cost of Part:** What you paid for the screen.
 - **Labour:** Your "skill fee." How much is your *time* and *expertise* worth?
- **Record Keeping:**
 - Get a simple "Job Book."
 - Write down *every* job:
 - Date, Client Name, Phone
 - Device Model, Fault
 - Part Cost (e.g., N15,000)
 - Total Paid (e.g., N20,000)
 - **Profit:** (e.g., N5,000)
 - At the end of the month, you can see if you are *actually* making money.

EVALUATION (WEEK 10)

1. What is a "niche," and why is it important for a new repair business?
2. What is the *single most important* ethical rule for a technician regarding a client's device?
3. What is the difference between "Cost of Part" and "Labour" in your pricing?
4. What is the *best* free way for new customers to find your shop?
5. Why is "word-of-mouth" so important?

ASSIGNMENT (WEEK 10)

1. Create a "startup tool list" for your new, small GSM repair shop.

2. List the 10 *most essential* tools you would buy first.
3. Next, create a "startup inventory list." List the 5 *most common* spare parts you would stock, based on the phones you see people using every day.



WEEK 11: REVISION

This week is for a comprehensive review of all topics covered in the Third Term.

- **Advanced PC Repairs (W1-2):**

- BIOS/UEFI: Updating (why/how/dangers).
- OS Install: Clean install process (backup, boot media, partition, drivers).
- Upgrades: RAM (DIMM/DDR), HDD->SSD (cloning vs. clean), GPU (PSU, size, slot).

- **PC Security (W3):**

- Malware: Virus, Worm, Trojan, Ransomware.
- Prevention: Legitimate software, updates, phishing awareness.
- Removal: Safe Mode, scanners (Malwarebytes), sfc, Factory Reset.

- **PC Networking (W4):**

- LAN/WLAN: Router, Switch, AP, SSID.
- IP: DHCP, ipconfig (169.254 is bad).
- Troubleshooting: ping gateway, ping 8.8.8.8 (internet), ping google.com (DNS).

- **GSM Software (W6, W8):**

- Optimization: Rooting (pros/cons), Custom ROMs (LineageOS, TWRP), OTA.
- Troubleshooting: The "ladder" (Force Restart -> Clear Cache -> Safe Mode -> Wipe Cache Partition -> Factory Reset).

- **GSM Connectivity (W9):**

- The "ladder" (Check bill -> Restart -> Check network type).
- **APN (Access Point Name):** The #1 technician fix for "no data."

- **Business (W10):**

- The Plan (Niche, Location), The Skills (Honesty, Data Privacy), The Money (Pricing, Records).